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Multi-target TME Reprogramming Awakens TLS driven Immunity in Ovarian Cancer

Wenju Peng[#], Yao Chen[#], Wenjing Jiang[#], Qiongli Yu, Xu Huang, Naiyu li, Wulin Shan, Wei Shen, Yuan Tian, Yuxuan Shi, Hanxue Sun, Huanhuan Mao, Haomin Zhang, Xiaofei Lu, Yumeng Wang, Wenjun Wu, Baoli Hong, Ting Han, Mengxi Tang, Feng Wu[*], Bairong Xia[*] & Wei Xiong[*]

Affiliations:

Division of Gynecologic Oncology in the Department of Obstetrics and Gynecology, The First Affiliated Hospital of USTC, Anhui Province Key Laboratory of Biomedical Imaging and Intelligent Processing of IAI, Division of Life Sciences and Medicine, University of Science and Technology of China, Department of Gynecologic Oncology, Anhui Provincial Cancer Hospital, Hefei, Anhui, 230001, China

[#]These authors contributed equally

[*] Correspondence: fengwu@ustc.edu.cn, xiabairong@ustc.edu.cn & wxiong@ustc.edu.cn

Summary

High-grade serous ovarian carcinoma (HGSOC) remains the deadliest gynecologic malignancy, with an immunosuppressive tumor microenvironment (TME) that has eluded therapeutic targeting. AI-driven screening identified surufatinib as a multi-target agent capable of co-inhibiting CSF1R, VEGFR and FGFR1 to dismantle this network. Combining surufatinib with the PARP inhibitor pamiparib as the PASSION regimen yielded synergistic antitumor and anti-ascites efficacy preclinically. In a phase II neoadjuvant trial for advanced unresectable HGSOC, PASSION achieved a 100% objective response rate and 92.3% R0 resection rate with manageable safety. Spatiotemporal multi-omics demonstrated that PASSION fundamentally reprograms the TME, inducing de novo B:T tertiary lymphoid structures (TLS) that convert immunologically "cold" tumors into "hot" ones, while concurrently upregulating PD-L1. Sequential anti-PD-1 blockade thereafter unleashed the full potential of TLS-educated immunity, notably sensitizing BRCA wild-type tumors otherwise refractory to immunotherapy. This study establishes AI-guided, multi-targeted TME reprogramming as a paradigm for in situ immune induction, offering a clinically validated strategy to overcome immune resistance in ovarian cancer.

Introduction

High-grade serous ovarian carcinoma (HGSOC) remains a profound clinical challenge, not for a lack of initial response, but for the near-inevitability of therapeutic resistance and recurrence (Kordowitzki et al., 2025). The limitations of conventional therapies have increasingly pointed to the tumor microenvironment (TME) as a decisive factor in treatment failure (Hansen et al., 2016). While poly (ADP-ribose) polymerase inhibitors (PARPi) have greatly improved outcomes for homologous recombination-deficient tumors, they

eventually stop working, not solely because of cancer cell-intrinsic survival mechanisms, but because the TME actively shields residual tumor cells from therapeutic eradication (Caruso et al., 2025). This view holds that the TME is not merely a passive backdrop but an active, co-evolving ecosystem that influences disease course and therapeutic outcomes. This is particularly relevant in HGSOC, where the immunosuppressive TME promotes immune escape, platinum resistance and peritoneal spread (Dagogo-Jack and Shaw, 2018).

In HGSOC, the TME forms a self-reinforcing, immunosuppressive network rather than a loose collection of cell types (Bejarano et al., 2021). Three stromal cell types including tumor-associated macrophages (TAMs), tumor endothelial cells (TECs) and cancer-associated fibroblasts (CAFs) are central to establishing a cold tumor phenotype (Tang et al., 2023). TAMs are maintained by CSF1R signaling and polarize toward an M2-like state. They secrete VEGF, which induces TECs to form dysfunctional, hyper-permeable vessels (Mantovani et al., 2017). These abnormal vessels fail to support T-cell infiltration and create hypoxic niches that reinforce TAM-driven immunosuppression (Zheng et al., 2025). CAFs further contribute to this environment by producing a dense extracellular matrix and upregulating immune checkpoint molecules (Zhang et al., 2023). This immunosuppressive milieu largely explains why immune checkpoint inhibitors (ICIs) have limited efficacy in unselected HGSOC: without pre-existing immune infiltration, PD-1/PD-L1 blockade cannot elicit a meaningful response (Zhao et al., 2025). Such reciprocal crosstalk also explains why single-agent strategies targeting the TME often fail. Inhibiting one pathway, such as VEGF, can trigger compensatory upregulation of another, like CSF1R, allowing the suppressive network to persist (Dalton et al., 2017b). Effective disruption of this network therefore requires simultaneous blockade of its core signaling nodes.

The critical role of these three stromal populations is well supported by a systematic evaluation of the existing literature. Field-wide research has converged on distinct receptor signaling pathways linked to each cell type: CSF1R is the most extensively studied target for TAM modulation (Truxova et al., 2023), featured in nearly half of relevant publications, while the VEGF/VEGFR axis—with FLT1 (VEGFR1) as a central mediator—dominates anti-angiogenic investigations aimed at TECs. For CAFs, FGFR family receptors represent the primary druggable candidates for functional regulation (Liu et al., 2026). Importantly, multiple independent clinical studies have demonstrated that high expression of CSF1R, VEGFR1, or FGFR correlates with poor overall survival in HGSOC patients, reinforcing their potential as therapeutic targets (Birrer et al., 2007; Dalton et al., 2017a, 2017b). Together, CSF1R, VEGFR, and FGFR signaling pathways constitute the core regulatory network of the immunosuppressive and protumoral TME in HGSOC, establishing a clear rationale for their simultaneous blockade to remodel the TAM, TEC, and CAF compartments (Kabir et al., 2026).

A systems-level strategy to reprogram this TME therefore requires an agent capable of concurrently inhibiting these master regulators. Here, we conceived such a strategy by deploying an artificial intelligence-driven platform to computationally screen for a single multi-kinase inhibitor with high predicted activity against CSF1R, VEGFR, and FGFR1. We rationalized that pairing TME remodeling through multi-targeted kinase inhibition with PARP inhibitor-induced cytotoxicity could generate synergistic antitumor activity. To

investigate the clinical feasibility and underlying mechanisms of this approach, the combination was evaluated in a neoadjuvant clinical trial for advanced, unresectable HGSOC. This trial design uniquely enabled the acquisition of paired pre- and on-treatment tumor specimens, which were subjected to integrative, spatiotemporal multi-omics profiling. Through this deep biological dissection, the study aimed to define how therapeutic modulation of the TME rewires the immune landscape and to identify vulnerabilities that could be exploited by rational immunotherapy combinations. By establishing a paradigm of proactive TME reprogramming, this work provides a clinically validated foundation for overcoming immune resistance and improving outcomes in this lethal malignancy.

Results

AI-Driven Discovery of Surufatinib as a Multi-Target TME Modulator

To discover a single agent capable of simultaneously inhibiting the core TME drivers, we developed a deep learning framework for the high-precision prediction of drug-target interactions (DTIs) **(Figure 1A)**. Our "Comprehensive Drug Evaluation System (CDES)" integrates pre-trained models for molecular representation: ChemBERTa generates embeddings from drug SMILES notations, while ESM2 encodes protein sequences (Chithrananda et al., 2020). These initial representations were further processed through dedicated feature refinement modules, using convolutional neural networks (CNNs) for proteins and graph convolutional networks (GCNs) for drugs, followed by a pairwise interaction module. A key innovation of CDES is the incorporation of a multi-channel bidirectional cross-attention mechanism, which dynamically models complex, non-linear relationships between drug and target features, in contrast to standard feature concatenation approaches. To enhance its relevance for oncology, the framework was retrained on a large, curated dataset of approximately one million high-quality DTI pairs enriched with cancer-related interactions. This domain-specific training improved the model's performance and generalization capability for oncology tasks compared to existing generalist models. For example, CDES assigned substantially higher confidence scores to known positive drug-target pairs, whereas models such as Boltz-2 frequently produced scores near the decision threshold, indicating improved discrimination by CDES (Passaro et al., 2025).

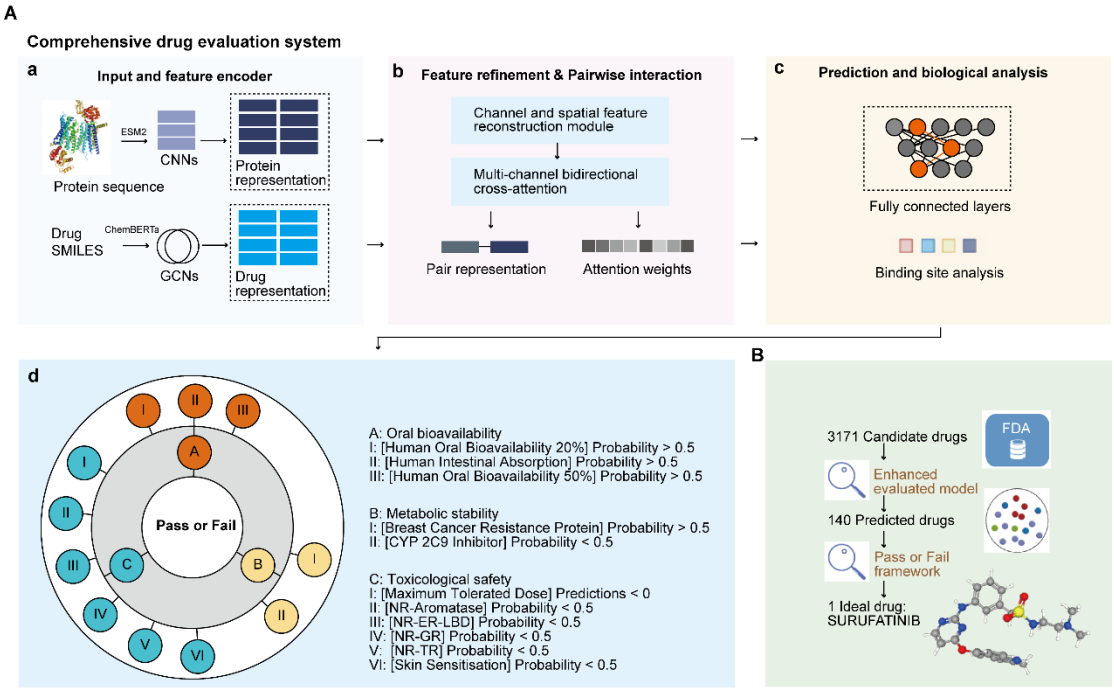


Figure 1 Comprehensive Framework for Oncology Drug Screening Identified Surufatinib as a Candidate.

(A) Architecture of the comprehensive deep learning framework for drug-target interaction (DTI) prediction. The model integrates pretrained molecular representations (ChemBERTa for drugs, ESM2 for proteins), refines features via dedicated modules (CNN for proteins, GCN for drugs), and employs a multi-channel bidirectional cross-attention mechanism to model complex interactions. It was retrained on a cancer-enriched DTI dataset to optimize performance for oncology applications. Schematic of the stringent "pass/fail" pharmacological screening funnel based on 11 CQAs. Candidates predicted to bind the five key TME targets were evaluated across three core pillars: Oral Bioavailability, Metabolic Stability, and Toxicological Safety. A compound had to satisfy all 11 criteria to advance.

(B) Identification of Surufatinib as the optimal candidate. Surufatinib successfully passed all stages of the integrated computational and in silico screening cascade. It is highlighted as the final output, demonstrating high predicted binding affinity to the TME targets alongside a balanced and favorable pharmacological profile for oral administration.

Using the CDES model, we screened a library of 3,171 FDA-approved or Phase III clinical-stage compounds against the five key TME targets (CSF1R, VEGFR1,2,3 and FGFR1). The analysis identified 140 candidates predicted to bind strongly (prediction probability > 0.9) and concurrently to all five targets. To evaluate their translational potential, we subjected this computationally prioritized list to a multi-stage pharmacological screening pipeline. We established a binary "pass/fail" framework based on 11 Critical Quality Attributes (CQAs) that collectively define an ideal oral oncology drug **(Figure 1A)**. These CQAs were chosen to assess three dimensions: Oral Bioavailability (e.g., predicted human oral absorption), Metabolic Stability (e.g., interaction profiles with enzymes like CYP450), and Toxicological Safety (e.g., predicted maximum tolerated dose and specific off-target liabilities). A compound was required to pass all 11 criteria to advance, ensuring

that final candidates were not only predicted to be potent multi-target binders but also exhibited a pharmacological and safety profile suitable for clinical development.

This integrated computational and in silico pharmacological screening cascade culminated in the identification of a single optimal candidate: Surufatinib (Syed, 2021). Surufatinib was the only compound to satisfy all 11 CQAs, reflecting a combination of predicted oral bioavailability, tumor accumulation potential, and an acceptable safety profile (**Figure 1B**). This data-driven discovery highlights Surufatinib's multi-faceted suitability and nominates it as a prime candidate for further mechanistic validation and therapeutic exploration against the high-risk TME in HGSOC.

Preclinical Efficacy of Surufatinib plus Pamiparib in HGSOC Mouse Models

We functionally validated our AI-driven prediction by testing surufatinib's capacity to remodel the TME *in vivo*, both alone and in combination with the PARP inhibitor pamiparib. To preserve the intact stromal-immune compartment essential for evaluating TME-targeted therapy, we used an immunocompetent, syngeneic mouse model bearing aggressive, metastatic BRCA-mutant (BRCAmut) ovarian tumors. (**Figure 2A**). Tumor-bearing mice were randomized into four treatment arms: vehicle, surufatinib, pamiparib, or the combination of both agents (PASSION regimen), with treatment initiated after a 4-week engraftment period (**Figure 2B**). Longitudinal bioluminescence imaging demonstrated that the combination of surufatinib and pamiparib (PASSION) produced synergistic tumor growth suppression, resulting in near-complete arrest of tumor progression and the lowest tumor burden at the study endpoint (**Figure 2C, D**). By contrast, neither monotherapy achieved comparable tumor control.

We next evaluated the impact on malignant ascites, a life-limiting complication of advanced HGSOC. Control and monotherapy-treated mice developed hemorrhagic ascites, a hallmark of advanced peritoneal disease (Kipps et al., 2013). In contrast, the PASSION regimen markedly reduced ascites formation, with a statistically significant reduction in ascites volume compared to all other groups (**Figure 2E**). Gross pathological examination of combination-treated mice showed less intestinal distension and reduced liver and spleen damage.

To determine whether this efficacy extended to the more therapeutically challenging BRCA wild-type (BRCAwt) setting, we replicated the study in a syngeneic BRCAwt model. The PASSION regimen achieved comparable tumor control and near-complete ascites suppression, independent of BRCA status (**Supplementary Figure 1**). These results suggest that the combination's activity does not depend solely on PARPi-mediated synthetic lethality and likely involves TME remodeling, providing a preclinical rationale for clinical evaluation in patients with advanced HGSOC.

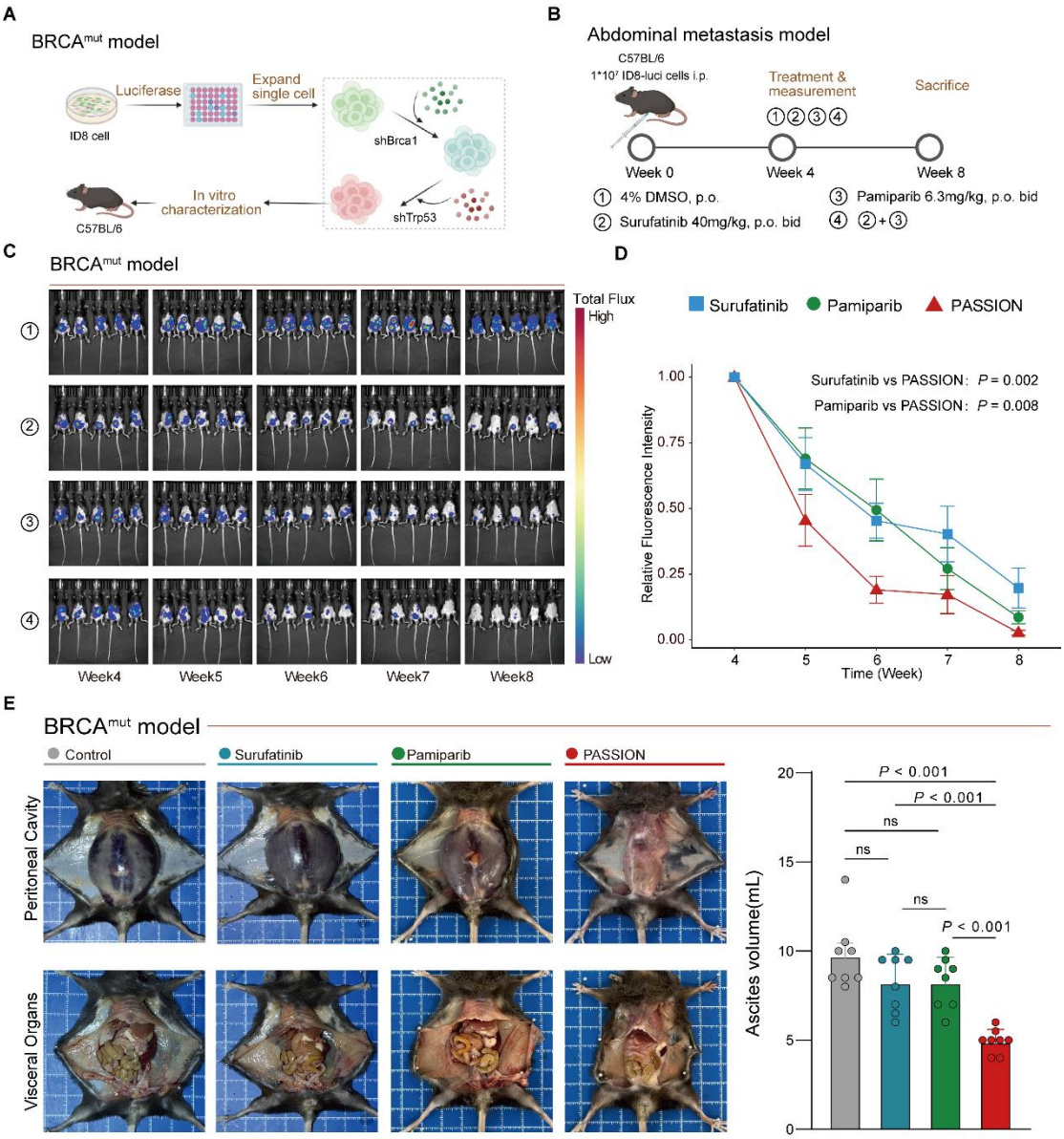


Figure 2 Efficacy of Pamiparib-Surufatinib Combination in Ovarian Cancer Animal Models.

(**A**) Diagrammatic representation of Trp53−/−Brca1−/− ID8 cells transplanted into mice.

(**B**) Diagrammatic representation of the animal experimental paradigm.

(**C**) The bioluminescent images of all mice in each respective group.

(**D**) The temporal changes in BRCAmut tumor burdens from week 4 to week 8 treatment period, as gauged by luminescence intensity, with treatment annotated by colors. Global intergroup differences were assessed using Bonferroni post-hoc tests following repeated-measures ANOVA.

(**E**) Ascites condition and anatomical diagrams of BRCAmut (left) model after treatment, with quantitative intergroup comparisons of ascites volume analyzed by unpaired two-tailed Student's t-test (right).

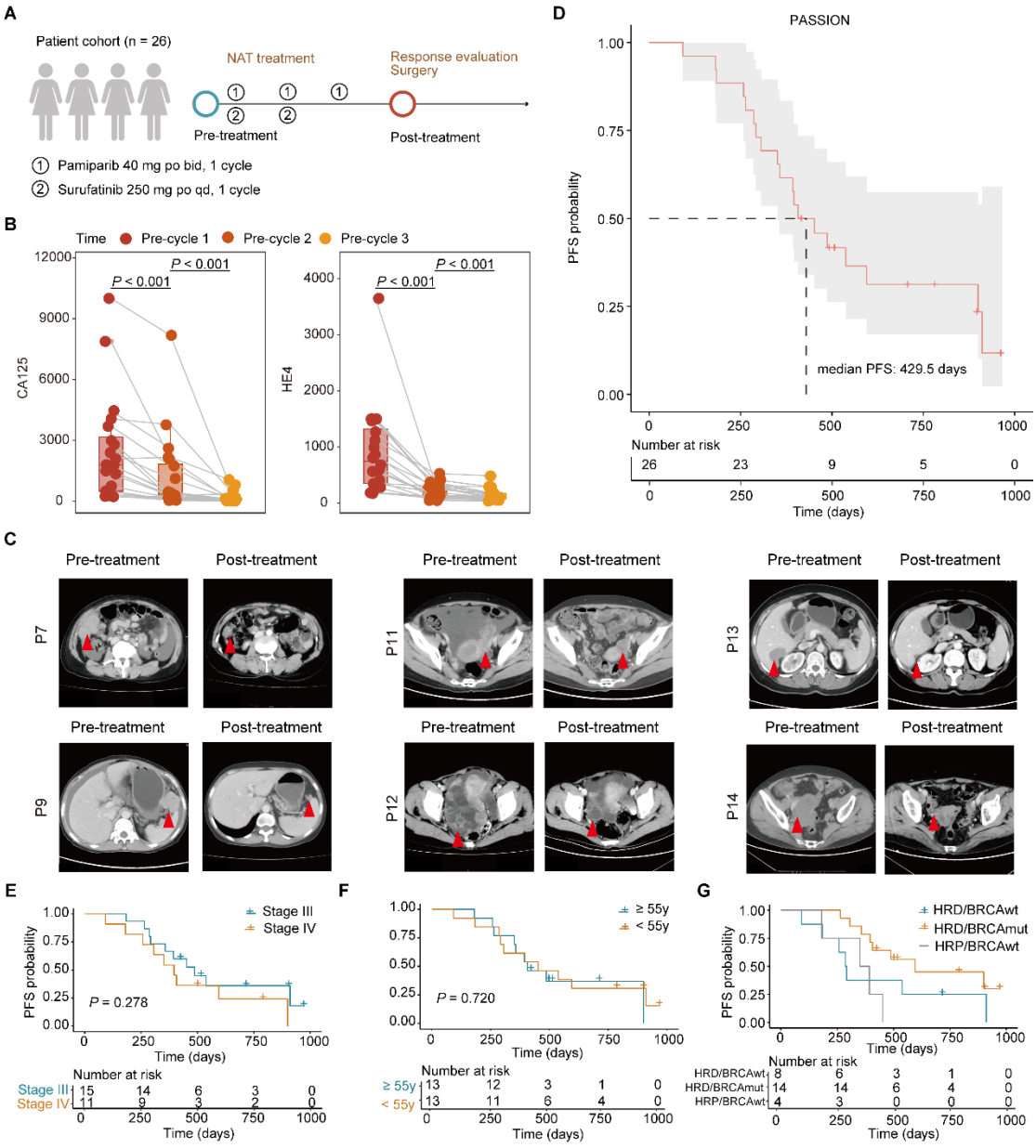


Figure 3 Pamiparib Plus Surufatinib as Neoadjuvant Treatment for Advanced Unresectable Ovarian Cancer (PASSION)

(**A**) Schematic of the prospective, single-arm, single-center phase II clinical trial design. Patients with advanced, unresectable HGSOC received three cycles of neoadjuvant therapy (NAT) with oral pamiparib and surufatinib, followed by interval debulking surgery (IDS).

(**B**) Longitudinal dynamics of serum tumor biomarkers CA125 and HE4 during NAT. Both markers showed a profound and statistically significant decline from baseline over the three treatment cycles ($p < 0.001$ for both), indicating a potent, early treatment response.

(**C**) Exemplary serial axial CT images from a representative patient demonstrating a partial response. Post-NAT imaging evaluation in the efficacy-evaluable population (n=26) revealed an objective response rate (ORR) of 100%. Subsequent IDS achieved an R0 resection rate of 92.3%, including one patient with a pathologic complete response (pCR).

(**D**) Kaplan-Meier curve for PFS in the PASSION cohort (n=26). The median PFS was 429.5 days.

(**E-G**) Kaplan-Meier curves for PFS in key patient subgroups. Subgroup analyses showed no significant

difference in PFS between patients with stage III and stage IV disease (**E**) or by age (**F**), indicating broad clinical benefit. As expected, BRCA mutation status remained a significant prognostic factor (**G**). P values were calculated using the log-rank test.

Phase II Neoadjuvant Trial of the PASSION Regimen

To determine whether the synergistic antitumor activity and ascites suppression observed in our preclinical models could translate to patients, we conducted a prospective, single-arm, single-center phase II clinical trial (the PASSION trial, NCT05652283) in patients with advanced, initially unresectable HGSOC. The neoadjuvant setting allowed paired pre- and post-treatment tissue acquisition, enabling downstream mechanistic dissection of TME remodeling. Eligible patients received three cycles of oral pamiparib and surufatinib, followed by interval debulking surgery (IDS) (**Figure 3A**).

Between November 2022 and November 2025, 32 patients were screened. Of the 31 who initiated treatment, 26 completed the planned neoadjuvant therapy (NAT) and underwent IDS, constituting the efficacy-evaluable population (median age 56 years; 10 patients with stage IV disease) (**sTable 1, Supplementary Figure 2**). Longitudinal monitoring of serum CA125 and HE4 revealed a statistically significant decline in both biomarkers as early as the first treatment cycle ($p < 0.001$), indicating rapid and potent biological activity (**Figure 3B**).

Radiological evaluation post-NAT demonstrated an objective response rate (ORR) of 100% (26/26), with serial CT imaging confirming substantial regression of target lesions (**Figure 3C, sTable 2**). This radiographic response translated into favorable surgical outcomes: an R0 resection rate of 92.3% (24/26) was achieved, and one patient attained a pathological complete response (pCR). These outcomes compare favorably with historical rates for platinum-taxane neoadjuvant chemotherapy, which report ORRs of 60–78% and R0 rates of 33–85% (Kehoe et al., 2015; Onda et al., 2016; Vergote et al., 2010). The markedly high R0 rate is consistent with the TME-modifying activity of the PASSION regimen, by alleviating tumor burden and stromal desmoplasia as observed preclinically, converts initially unresectable lesions into operable disease.

At the time of analysis, median PFS was 429.5 days, with median OS unreached (**Figure 3D**). Subgroup analyses demonstrated that the PFS benefit was broad and consistent, with no significant differences observed between patients with stage III versus stage IV disease, a notable finding given the poor prognosis of stage IV HGSOC, or across age groups (Cai et al., 2024; Jakob et al., 2023) (**Figure 3E, F**). As anticipated, BRCA-mutant status was associated with a trend toward longer PFS (Petrillo et al., 2017). Importantly, however, substantial benefit was also evident in the BRCA wild-type subgroup (**Figure 3G**), consistent with the syngeneic model results and suggesting that TME remodeling contributes to the therapeutic effect, rather than BRCA-dependent synthetic lethality alone. The full analysis set (FAS) consisted of 32 patients with a median age of 56.00 years (49.25–61.00). Among them, 20 patients (62.5%) had stage III disease and 12 patients (37.5%) had stage IV disease. The median PFS was 452 days, and the 12-month PFS rate was 12/32, while median OS remained unreached (**Supplementary Figure 3**). Consistent with the above surgery subgroup findings, broad and consistent PFS benefit was observed across the overall FAS population.

The safety profile of the PASSION regimen was manageable in the FAS population (n=32), with most treatment-related adverse events (TRAEs) classified as Grade 1–2. Hematologic toxicities were the most frequent TRAEs: decreased white blood cell count (12 patients, 37.5%), neutropenia (9 patients, 28.1%), anemia (11 patients, 34.4%), and thrombocytopenia (5 patients, 15.6%). Other common adverse events included urine protein elevation (15 patients, 46.9%), elevated AST (1 patient, 3.1%), elevated ALT (1 patient, 3.1%), vomiting (1 patient with Grade ⩾3, 3.1%), and laxness (1 patient with Grade ⩾3, 3.1%). No hand-foot syndrome was reported in any patient. Three patients (9.4%) discontinued treatment due to adverse events; no patients experienced AE-related death or dose reduction. Grade ⩾3 adverse events were predominantly hematologic toxicities, and no Grade 4 adverse events were observed.

These findings demonstrate that the PASSION regimen yields high response and R0 resection rates in advanced HGSOC. The availability of paired pre- and post-treatment specimens enabled subsequent multi-omics profiling to examine the mechanisms of TME remodeling associated with these responses.

Multi-Omics Census Reveals Therapy-Induced Lymphoid Enrichment

To investigate the mechanisms underlying the clinical activity of the PASSION regimen, we performed integrated multi-omics profiling on paired peritoneal and omental metastasis specimens obtained before and after NAT (**Figure 4A, sTable 3**). This strategy combined bulk transcriptomics, proteomics, single-nucleus RNA sequencing (snRNA-seq), and spatial transcriptomics to characterize TME remodeling at various resolution. After rigorous quality control, we profiled a total of 33 transcriptomes (14 patient-matched pairs), 35 proteomes (15 matched pairs), and 21 snRNA-seq libraries (6 matched pairs), alongside 8 spatially resolved transcriptomic maps selected to enable matched comparisons.

Integrative snRNA-seq analysis using Harmony-batch-corrected data from 235,196 high-quality nuclei revealed the cellular architecture of the HGSOC TME (**Figure 4B**). Unsupervised clustering defined seven major lineages: epithelial cells (the malignant compartment), T/NK cells and B/plasma cells (the adaptive immune axis), and myeloid, stromal, and endothelial cells (collectively forming the immunosuppressive compartment). These identities were validated by the localized expression of canonical markers including EPCAM, CD3D, MS4A1 and JCHAIN (**Figure 4B-C**; **Supplementary Figure 4**).

We next quantified therapy-induced compositional shifts using the Ro/e index to assess the sampling-time preference of each lineage. This global analysis revealed a consistent shift: post-treatment samples from the overall cohort exhibited a marked enrichment of lymphoid cells, with T/NK and B cells shifting from a pre-treatment distribution deficit (Ro/e: 0.64 and 0.36, respectively) to a clear post-treatment preference (1.26 and 1.46). In contrast, the relative abundance of malignant epithelial cells declined (Ro/e: 1.10 to 0.92), while the stromal, endothelial, and myeloid compartments showed less pronounced changes (**Figure 4D**). These results indicate that the PASSION regimen shifts the TME from an epithelial-stromal-dominated state toward a lymphoid-inflamed phenotype, consistent with conversion from “cold” to “hot”. We next assessed whether these changes were associated with clinical outcomes and which immune subsets were involved.

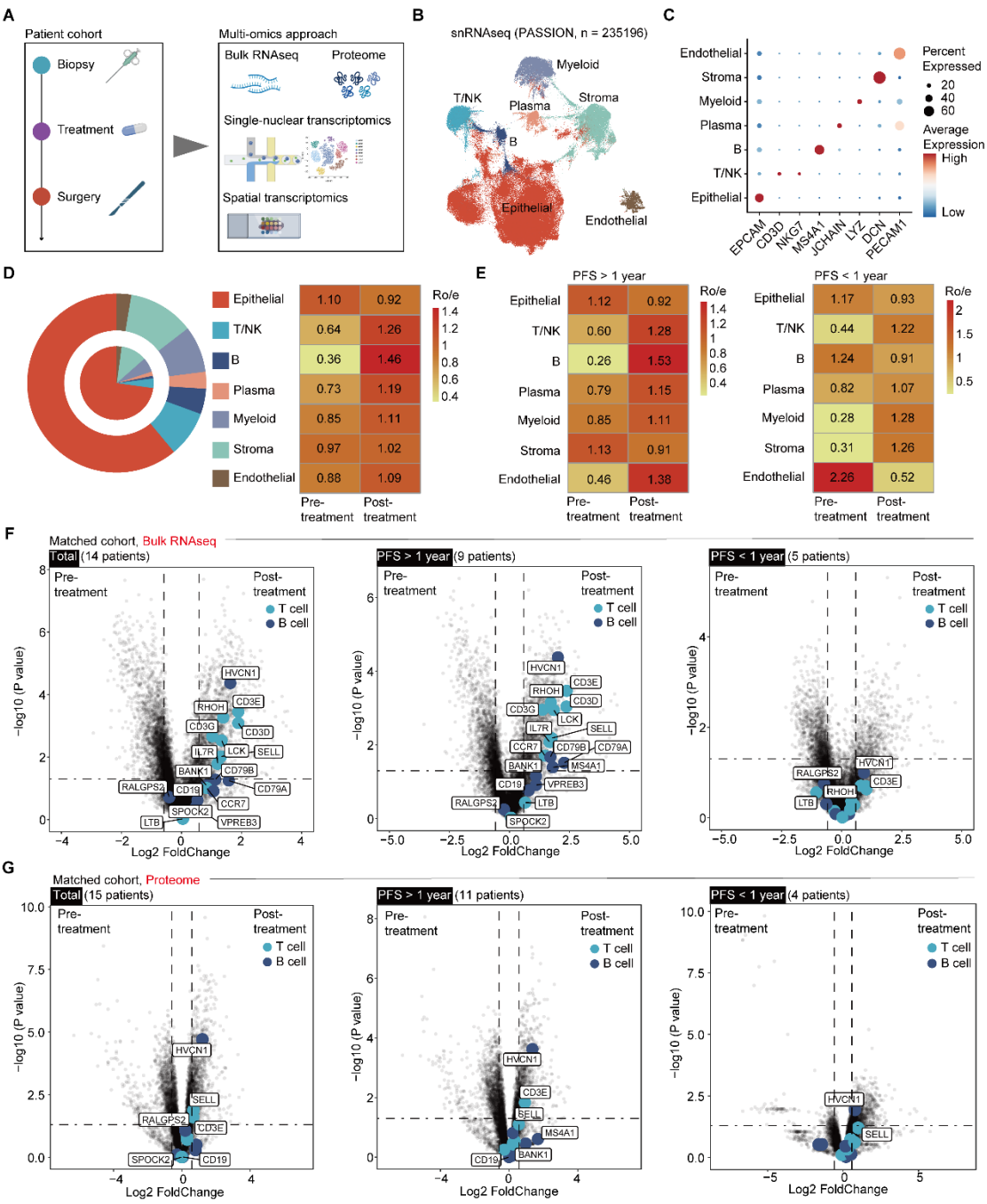


Figure 4 Multi-omic Profiling Reveals Immune Remodeling in Response to PASSION Therapy.

(**A**) Schematic overview of the clinical cohort design and multi-omics data collection, including transcriptomics, proteomics, snRNA-seq, and spatial profiling.

(**B**) UMAP visualization of snRNA-seq data from tumor samples, with major cell types annotated by distinct colors.

(**C**) Dot plot showing the expression of canonical marker genes across major cell type.

(**D-E**) Nested pie-donut charts illustrating the distribution of major cell types before (inner) and after (outer) treatment (left) and heatmap of Ro/e values illustrating shifts in major cell type composition across NAT treatment (right) of total patients (**D**) and those with PFS > 1 year and PFS < 1 year (**E**).

(**F**) Volcano plot depicting differentially expressed genes between pre- and post-treatment samples in

sample-matched patients (left) , patients with PFS > 1 year (middle) and PFS < 1 year (right), T cell– and B cell–related genes are highlighted in color.

(**G**) Volcano plot of proteomic data showing differentially expressed proteins between pre- and post-treatment samples in sample-matched patients (left) , patients with PFS > 1 year (middle) and PFS < 1 year (right), with T cell– and B cell–related proteins annotated by color.

Clinical Outcome-Linked Reprogramming of the TME

We next asked whether the global lymphoid enrichment observed following PASSION therapy translated into clinical benefit. Stratifying patients by PFS (>1 year vs. <1 year) revealed a clear divergence: the post-treatment enrichment of T/NK and B cells, as quantified by the Ro/e index, was enriched in patients with PFS >1 year and not observed in those with PFS <1 year (**Figure 4E, sTable 3**). This dichotomy was comprehensively validated in paired bulk transcriptomic and proteomic analyses. Patients with PFS >1 year exhibited an upregulation of T cell (e.g., CD3D, CD3E, LCK, IL7R) and B cell (e.g., CD19, CD79B, HVCN1) transcriptional programs post-treatment, a signature confirmed at the protein level (**Figure 4F-G, sTable 4-9**). In contrast, the PFS <1 year subgroup showed no significant lymphoid transcriptional activation.

Critically, this divergent immune remodeling was not a reflection of pre-existing differences in baseline immune contexture. Comparative analysis of pre-treatment transcriptomes and proteomes revealed no significant differences in any immune-related genes or proteins between the two outcome groups (**Supplementary Figure 5, sTable 10-11**). These results suggest that the clinical activity of the PASSION regimen is not determined by baseline immune contexture, but rather by its ability to induce an adaptive immune response.

To define the functional consequences of this divergence, we performed Gene Ontology (GO) enrichment analysis. The results revealed a major shift in transcriptional programs (**Supplementary Figure 6, 7**). In patients with PFS >1 year, PASSION therapy drove a transcriptional shift: pre-treatment programs dominated by cell cycle and proliferation were replaced by immune activation pathways, including leukocyte migration and cytokine-mediated signaling. By contrast, tumors from patients with PFS <1 year failed to engage this immune switch. Instead, their post-treatment transcriptional landscape was characterized by persistent or enhanced signatures of cell adhesion, extracellular matrix interaction and cellular migration, a profile suggestive of a fibrotic stromal response that may limit anti-tumor immunity. Thus, the favorable-outcome group showed a post-treatment transition from proliferation to immune activation, while the early-progression group maintained a stromal and adhesion-related expression profile.

Network-Level Rewiring of Intercellular Communication

To understand the molecular basis of these divergent clinical outcomes, we next mapped therapy-induced changes in intercellular communication networks using CellChat (Jin et al., 2021). Across the entire cohort, PASSION therapy fundamentally simplified the TME signaling landscape, drastically reducing both the number and overall strength of inferred cellular interactions (**Supplementary Figure 8a, b**). A broad array of oncogenic and stromal pathways, including MIF, EGF and ACTIVIN, were suppressed, and the intended

therapeutic targets, the VEGF, FGF and CSF signaling axes, were all significantly attenuated post-treatment, confirming effective pharmacological engagement of the core TME hubs (**Supplementary Figure 8C–F**).

However, this global network simplification did not uniformly translate into clinical benefit. We therefore examined whether the three targeted pathways were differentially modulated in patients with PFS >1 year versus <1 year. Stratifying by PFS (>1 year vs. <1 year) revealed clear mechanistic differences (**Supplementary Figure 9A, B**). For the VEGF axis, angiogenic receptors FLT1 and KDR were downregulated in endothelial cells of responders, but remained elevated or increased in the PFS <1 year group, suggesting persistent angiogenic signaling. More striking was the behavior of the CSF1/CSF1R axis: in patients with PFS >1 year, CSF1R expression was effectively suppressed, consistent with an attenuated immunosuppressive myeloid program; in the PFS <1 year group, however, this pathway was paradoxically upregulated, indicating persistent TAM-polarizing signaling, which may contribute to the stromal and immune-excluded phenotype.

The most unexpected insight emerged from the FGF pathway. Contrary to canonical literature assigning FGFR1 and FGFR2 to epithelial and endothelial compartments (Brewer et al., 2016), our snRNA-seq data revealed that these receptors were predominantly expressed by intratumoral B cells (**Supplementary Figure 8E**). This finding uncovers FGFR1 as a novel, non-canonical signaling node on B cells within the TME, identifying a previously unrecognized cellular target for surufatinib's third major activity. While both groups exhibited a post-treatment decrease in FGFR1 expression, this discovery directly implicates FGF pathway modulation in B cell biology and raises the question of how this transcriptional reprogramming impacts specific B cell subsets.

Collectively, these data demonstrate that clinical efficacy is not defined by the inhibition of any single target, but by the successful, coordinated co-suppression of three interconnected signaling circuits: VEGF-driven angiogenesis, CSF1R-mediated myeloid suppression, and, unexpectedly, FGFR signaling in intratumoral B cells. This multi-nodal network disruption, achieved selectively in PFS >1 year group, mechanistically explains the TME conversion from an immuno-suppressive to an immuno-active state, and directs our focus toward a deeper dissection of the B cell compartment.

Functional Dichotomy of B Cell Subsets and TLS Priming

Having established that clinical benefit depends on coordinated multi-nodal network disruption, we returned to the upstream question of what drives the divergent lymphoid remodeling. Since epithelial cells constitute the dominant compartment in HGSOC and can influence the immune environment, we asked whether therapy-induced epithelial reprogramming could influence B cell biology. Using hdWGCNA (Morabito et al., 2023), we constructed transcriptional networks in epithelial cells and identified 24 gene modules (M1-M24) representing diverse functional states (**Supplementary Figure 10 A-C, sTable 12**). Differential module eigengene (DME) analysis revealed a clear directional shift: Epi-M4 was down-regulated and Epi-M8 was up-regulated following treatment, most pronounced in the PFS > 1 year subgroup (**Supplementary Figure 10 D-F**). Functional enrichment linked these modules to B cell biology, Epi-M4 was enriched for genes regulating B cell apoptosis, while Epi-M8 was associated with marginal zone B cell differentiation

(**Supplementary Figure 10 G-H**). These results suggested that epithelial reprogramming might influence B cell fate, prompting a more detailed analysis of the B cell compartment.

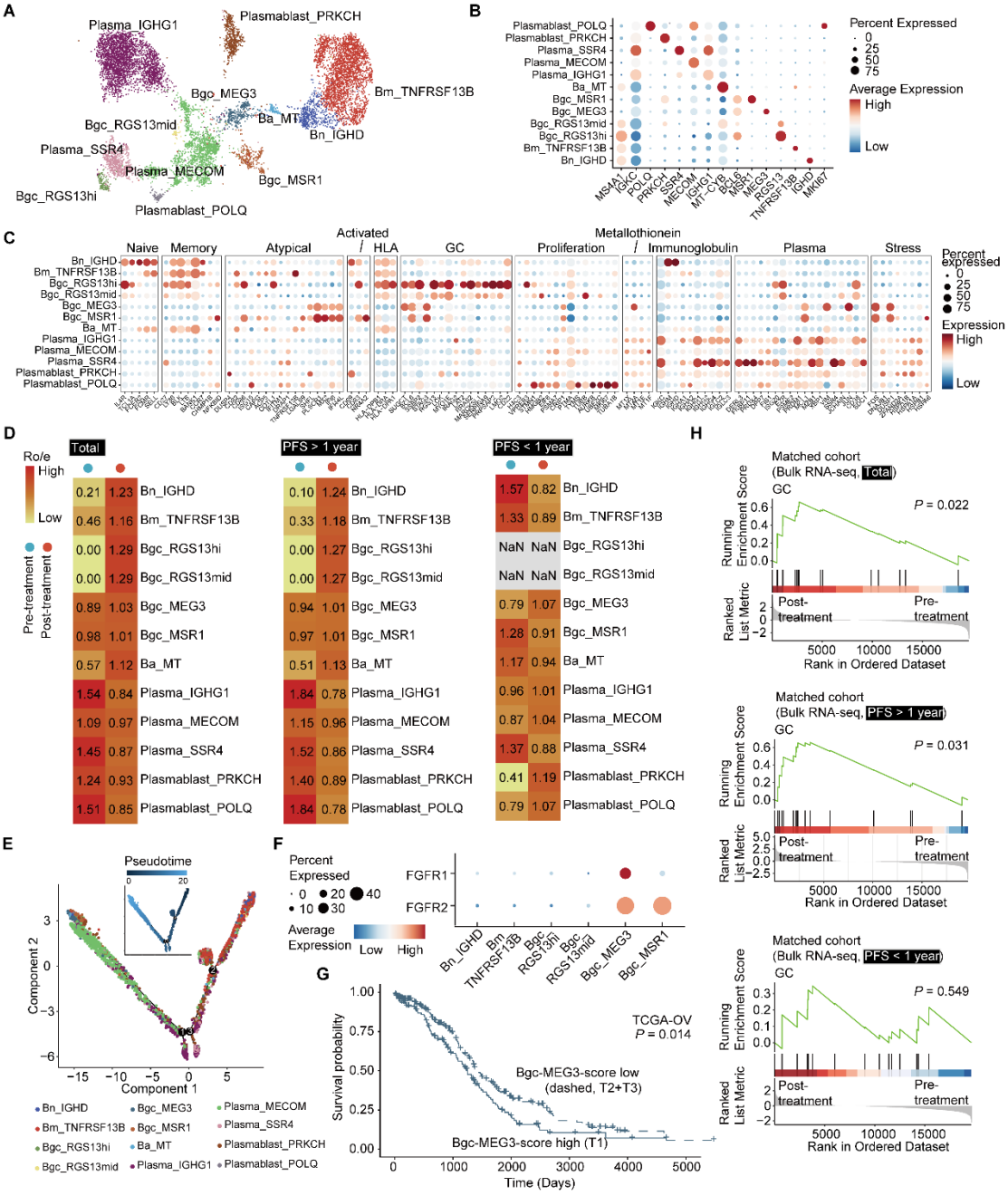


Figure 5 Subset Analysis Decodes B Cell and Plasma Cell Dynamics during PASSION Therapy

(A) UMAP visualization of B cell and plasma cell subclusters.

(B) Dot plot showing the percentage of cells expressing canonical marker genes (size) and mean expression levels (color) across B cell subclusters.

(C) Dot plot showing the percentage of cells expressing specific markers (size) and mean expression levels (color) across B cell subclusters.

(D) Heatmap of Ro/e values illustrating shifts in B cell subclusters composition across NAT treatment of total patients (left) and those with PFS > 1 year (middle) and PFS < 1 year (right).

(E) Monocle2 trajectory analysis showing developmental trajectory across B cell and plasma cell subclusters, with cells colored by corresponding pseudotime (up) and subclusters (bottom).
(F) Dot plot showing the percentage of cells expressing FGF receptors (size) and mean expression levels (color) across B cell subclusters.
(G) The Kaplan-Meier curves comparing the probability of survival according to Bgc-MEG3 score in TCGA-OV cohort. Patients were dichotomized into two groups based on the score distribution: the high-score group comprised the top one-third of patients with the highest scores, while the low-score group included the remaining two-thirds. P values were calculated with log-rank test.
(H) GC signature enrichment in post-treatment samples compared to pre-treatment samples of total patients (up) and those with PFS > 1 year (middle) and PFS < 1 year (bottom).

We therefore performed high-resolution analysis of B cells and plasma cells using our snRNA-seq dataset. Unsupervised clustering identified 12 transcriptionally distinct subclusters (**Figure 5A**, **Supplementary Figure 11**). Annotation using canonical markers (MS4A1/CD20, IGKC) and a curated functional gene panel defined specific phenotypes for each subcluster (Yang et al., 2024) (**Figure 5B, C**). Four clusters were identified as germinal center B cells (Bgc) based on BCL6 expression (Li et al., 2024). These Bgc clusters displayed substantial heterogeneity: Bgc_RGS13hi and Bgc_RGS13mid expressed canonical germinal center programs, whereas Bgc_MEG3 and Bgc_MSR1 lacked core germinal center markers (RGS13, CD22) and instead expressed atypical genes such as MX2 and IFI6, suggestive of an altered developmental path (**Figure 5C**).

We next investigated the dynamic changes of these subclusters in relation to treatment response. Specific subsets, Bn_IHGD, Bm_TNFRSF13B, Bgc_RGS13hi, and Bgc_RGSmid, showed preferential distribution in post-treatment samples (**Figure 5D**), an enrichment concentrated in the PFS > 1 year subgroup. Quantitative analysis confirmed their significantly higher abundance in the PFS > 1 year subgroup (**Figure 5D**). Bgc_RGS13hi and Bgc_RGSmid were virtually undetectable in the PFS < 1 year subgroup, revealing a stark dichotomy coupled to clinical benefit.

Functional analysis revealed that post-treatment samples broadly upregulated B cell proliferation, BCR signaling and B cell activation pathways (**Supplementary Figure 12A**). However, this activation was not generalized: key BCR signaling processes were predominantly enriched in the subsets expanded in PFS >1 year group, Bn_IHGD, Bm_TNFRSF13B, Bgc_RGS13hi, and Bgc_RGS13mid. In contrast, Bgc_MEG3 and Bgc_MSR1 failed to upregulate these canonical programs, instead exhibiting a divergent profile characterized by monocyte activation and chemotaxis pathways (**Supplementary Figure 12B**). These data indicate that PASSION therapy induces specific, functional activation of the BCR axis in a defined subset of B cells poised for productive anti-tumor immunity, whereas the Bgc_MEG3/MSR1 subsets represent an aberrant, non-protective state that may undermine beneficial immune remodeling.

Pseudotime trajectory analysis reconstructed a developmental continuum from naïve B cells (Bn_IGHD) through germinal center populations to terminally differentiated plasma cells (**Figure 5E**, **Supplementary Figure 13A, B**). Favorable-outcome-associated subsets mapped to early-to-mid pseudotime stages, while Bgc_MEG3 and Bgc_MSR1 localized to late pseudotime (**Supplementary Figure 13B, C**). Post-treatment trajectories from the

PFS > 1 year subgroup contained Bgc_RGS13hi and Bgc_RGS13mid populations, whereas these subsets were absent from PFS < 1 year trajectories (**Supplementary Figure 13C**). Bgc_MEG3 uniquely expressed high levels of FGF pathway receptors FGFR1 and FGFR2 (**Figure 5F**), linking surufatinib's FGF-inhibitory activity to this specific B cell subset.

GSEA confirmed that Bgc_MEG3 significantly upregulated monocyte activation and chemotaxis pathways relative to Bgc_RGS13hi/RGS13mid, aligning with a potential role in recruiting immunosuppressive myeloid cells (**Supplementary Figure 14A, B**). The clinical relevance of these subsets was underscored by their differential enrichment: Bgc_RGS13hi/RGSmid were selectively enriched in post-treatment samples from PFS > 1 year patients (**Supplementary Figure 14C, D; sTable 13-14**), and independent analysis of the TCGA-OV cohort revealed that lower Bgc_MEG3 transcriptional scores were significantly associated with improved survival (**Figure 5G, sTable 15**), validating its negative prognostic impact.

Given that productive intratumoral B cell responses are often coordinated within TLS featuring germinal center-like architecture (Yang et al., 2024), we assessed germinal center signature enrichment. Robust enrichment was evident exclusively in post-treatment samples from the PFS > 1 year subgroup (**Figure 5H**), directly linking the expansion of functional Bgc_RGS13hi/RGSmid subsets to the formation or enhancement of TLS/GC-like niches, a hallmark of effective anti-tumor immunity that shifts our focus to the spatial organization of these immune structures.

Collectively, these data demonstrate a critical functional dichotomy among B cell subsets driven by PASSION therapy. The regimen promotes expansion of Bgc_RGS13hi and Bgc_RGSmid subsets associated with productive germinal center reactions and active BCR signaling, while failing to restrain the FGFR1/2-expressing Bgc_MEG3 subset, which exhibits a monocyte-skewed, non-canonical functional profile. This bifurcation provides a mechanistic explanation for the divergent clinical outcomes between PFS subgroups and underscores that a successful germinal center-dependent humoral response is a key correlate of treatment efficacy, setting the stage for spatial dissection of the TLS niches in which these responses unfold.

Endothelial Reprogramming and Vascular Normalization

Given that VEGFR inhibition is a central pharmacological action of surufatinib (Syed, 2021), we next examined therapy-induced reprogramming of the tumor vasculature through high-resolution subclustering of the endothelial compartment. This analysis revealed six transcriptionally distinct subclusters: a lymphatic EC subset (Endo_Lymphatic_PROX1), three blood vascular EC subsets (arterial, tip cell, and venous), a hypoxic EC subset (Endo_Hypoxia_MT2A) marked by metallothionein expression, and a stress-activated subset (Endo_Stress_KAZN) (**Figure 6A-C**, **Supplementary Figure 15A**). The lymphatic endothelial subset exhibited a post-treatment enrichment, concentrated in the PFS > 1 year subgroup (**Figure 6D**), suggesting a therapy-induced shift toward vascular normalization associated with clinical benefit.

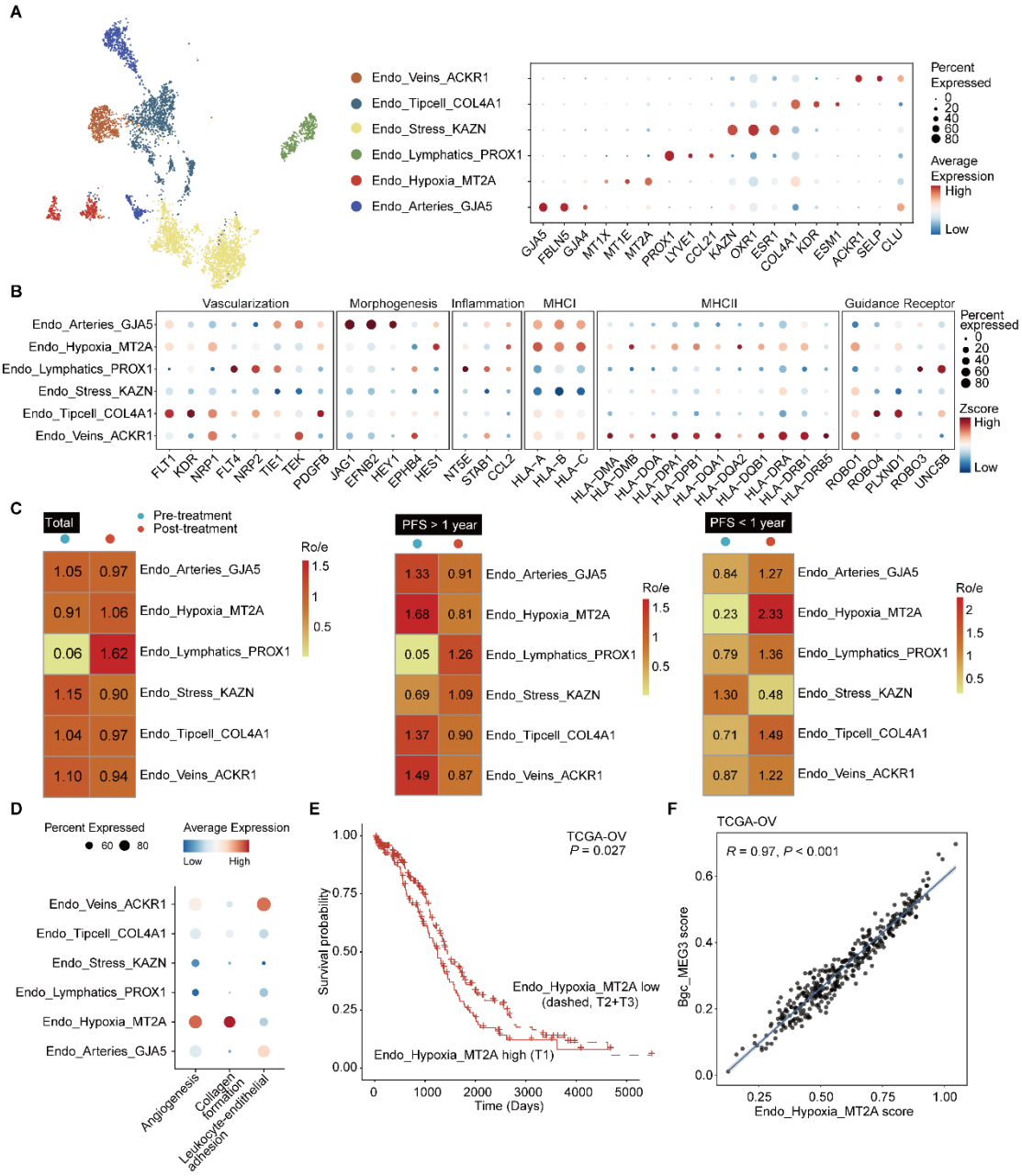


Figure 6 Subset Analysis Decodes Endothelial Cell Dynamics during PASSION Therapy.

(**A**) UMAP visualization of endothelial subclusters (left). Dot plot showing the percentage of cells expressing canonical marker genes (size) and mean expression levels (color) across endothelial subclusters (right).

(**B**) Dot plot showing the percentage of cells expressing functional genes (size) and mean expression levels (color) across endothelial subclusters.

(**C**) Heatmap of Ro/e values illustrating shifts in endothelial subclusters composition across NAT treatment.

(D) Dotplot comparing the functional scores of endothelial subclusters.

(E) The Kaplan-Meier curves comparing the probability of survival according to Endo-Hypoxia-MT2A score in TCGA-OV cohort. Patients were dichotomized into two groups based on the score distribution: the high-score group comprised the top one-third of patients with the highest scores, while the low-score group included the remaining two-thirds. P values were calculated with log-rank test.

(F) Scatter plot showing Pearson's correlation analysis of Bgc-MEG3 score versus Endo-Hypoxia-MT2A score based on the TCGA-OV ovarian cancer cohort; correlation coefficient and P value are annotated within the plot.

However, systematic scoring of vascular functional axes uncovered an unexpected finding: the Endo_Hypoxia_MT2A subcluster, rather than tip cells, exhibited the highest activity scores for both angiogenesis and collagen formation (**Supplementary Figure 15B**), correlated with inferior clinical outcomes (**Figure 6E, Supplementary Figure 15C-D**) and significantly elevated "regulation of vasculature development" pathway activity (**Supplementary Figure 15E**). This pro-angiogenic state in Endo_Hypoxia_MT2A occurred in the absence of elevated classical angiogenic RTKs—FLT1 (VEGFR1), KDR (VEGFR2), and NRP1 (**Figure 6B**), and association with Bgc-MEG3 (**Figure 6F, Supplementary Figure 15F**), suggesting that its dysregulated function is driven by alternative, non-canonical signaling pathways. This may represent a mechanism by which pathological angiogenesis persists after VEGFR inhibition.

In summary, the PFS >1 year subgroup showed expansion of lymphatic ECs and reduced tip cell angiogenic activity, whereas the PFS <1 year subgroup retained a hypoxia-driven angiogenic program independent of classical VEGFR signaling. This vascular resistance mechanism, together with CSF1R upregulation in myeloid cells and persistence of FGFR1/2-expressing Bgc_MEG3 cells in the PFS <1 year subgroup, suggested multiple resistance nodes. Whether this vascular remodeling facilitates or impedes the lymphoid infiltration and TLS formation observed in PFS > 1 year subgroup demands spatial interrogation, which we next pursued.

Spatial Mapping Reveals in situ TLS Neogenesis

While single-nucleus and bulk analyses quantify gene expression, they do not capture how immune cells are organized within the tumor (Yue et al., 2023). We therefore performed spatial transcriptomics on paired pre- and post-treatment specimens and integrated the data with an external scRNA-seq reference atlas (**Figure 7A**). Using CARD deconvolution (Tanevski et al., 2022), we mapped cell-type proportions onto each spatially resolved spot across tissue sections, transitioning from compositional enumeration to spatially resolved interrogation of lymphocyte organization (**Supplementary Figure 16-18**).

We first examined therapy-induced rewiring of spatial communication networks using mistyR, which models cell-population influences across spatial scales through complementary intraview (within-spot) and paraview (between-spot) models (**Supplementary Figure 19**). Post-treatment, the intraview model revealed a significant disruption of epithelial-to-myeloid spatial influence, a potentially tumor-promoting axis, while B cell regulatory influence on T cells was substantially enhanced, indicating the emergence of lymphocyte-coordinating hubs within local microenvironments (**Supplementary Figure 19A-B**). The paraview model captured broader contextual shifts: epithelial-myeloid coordination persisted across adjacent niches, while a novel, enhanced spatial relationship between epithelial cells and plasma cells emerged post-treatment, suggesting treatment-facilitated recruitment of antibody-secreting cells to the tumor periphery (**Supplementary Figure 19C-D**). These data demonstrate that PASSION

therapy orchestrates multi-scale spatial reorganization, attenuating pro-tumorigenic interactions while fostering immune-potentiating connections.

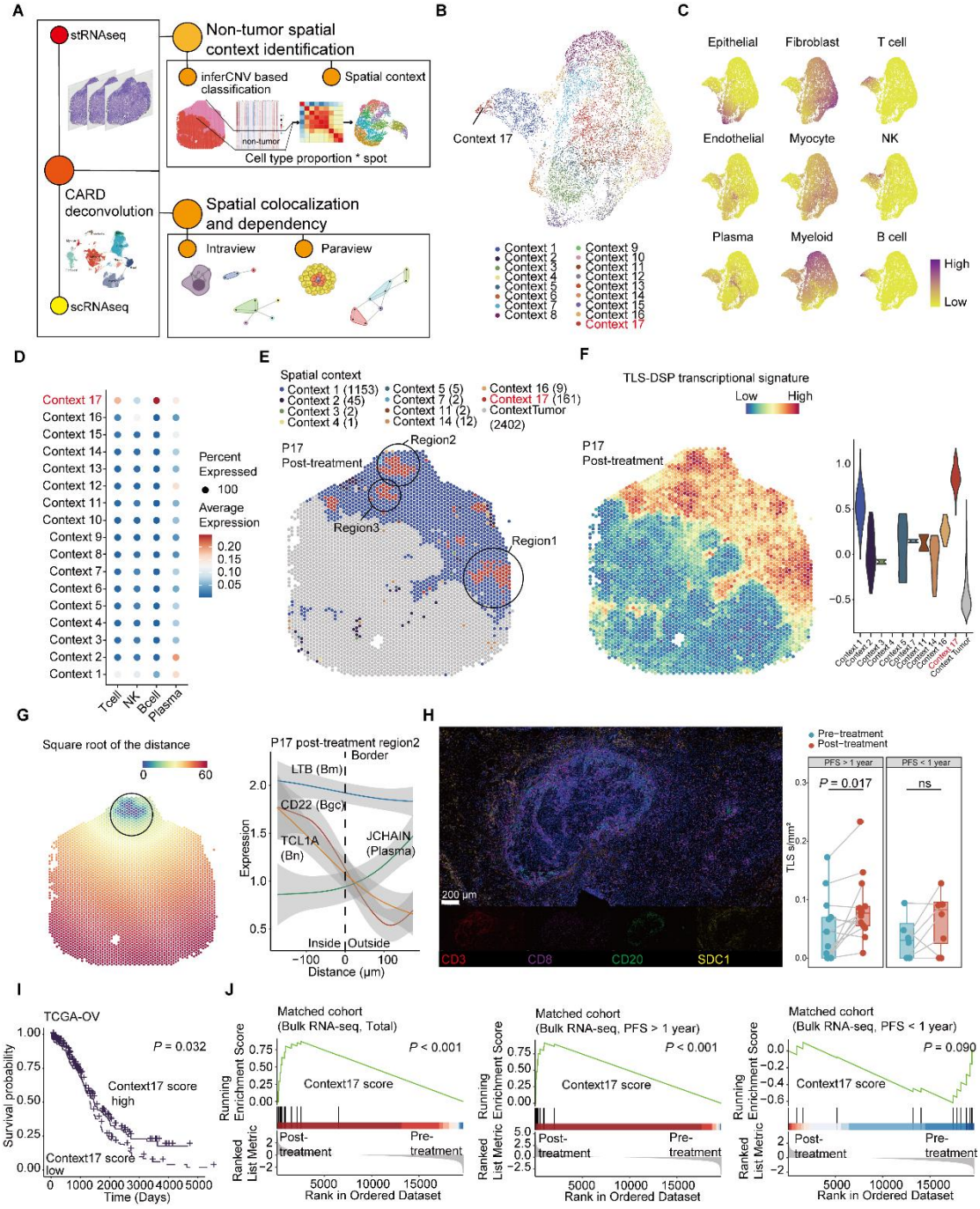


Figure7 PASSION Therapy Reshapes the Spatial Context of the Tumor Microenvironment, Enhancing Local Lymphocyte Interactions.

(**A**) Schematic illustration of spatial context identification.

(**B**) UMAP visualization of stRNA-seq data, with spatial context annotated by distinct colors.

(**C**) UMAP visualization of stRNA-seq data, colored by cell type proportion.

(**D**) Dot plot showing the proportion of major cell types across spatial contexts.

(**E**) Spatial distribution of context in post-treatment P17 sample.

(**F**) Spatial distribution of TLS-DSP signature in post-treatment P17 sample (left) and across spatial

contexts (right).
(**G**) spatial distribution of distance from region (left) and spatial expression pattern of B cell major lineages specific genes in region of post-treatment P17 sample (right).
(**H**) Representative mIHC images for CD3, CD8, CD20 and SDC1 from post-treatment patients. Scale bars 200μm (left). Box plot showing density of TLS aggregates within pre- and post-treatment samples split by PFS > 1 year and PFS < 1 year. Box plots: lower quartile, median, upper quartile (right).
(**I**) The Kaplan-Meier curves comparing the probability of survival according to Context 17 score in TCGA-OV cohort. Patients are dichotomized as high or low score using the median Context 16 score as cutoff. P values were calculated with log-rank test.
(**J**) Context 17 score enrichment in post-treatment samples compared to pre-treatment samples of all sample-matched patients (left), patients with PFS > 1 year (middle) and PFS < 1 year (right).

To identify functionally relevant immune niches, we computationally segregated malignant (CNV-positive) from non-malignant (CNV-negative) tissue regions using inferCNV, then performed unsupervised clustering of CNV-negative spots based on CARD-derived cell-type compositions (**Supplementary Figure 21**). This identified 17 distinct non-tumor spatial contexts (**Figure 7B**). Context 17 showed co-enrichment of B and T lymphocytes, unlike other contexts, suggesting adaptive immune interaction consistent with nascent TLS (**Figure 7C-D**). Context 17 was predominantly and specifically enriched in post-treatment samples from patients P14 and P17 (**Figure 7E**, **Supplementary Figure 21**). Both remained recurrence-free beyond 700 days of follow-up. In contrast, patients P18 and P20, whose post-treatment samples lacked Context 17, recurred at 396 and 92 days, respectively.

To determine whether Context 17 represents genuine TLS, we aligned our spatial maps with the established TLS-DSP transcriptional signature defined by MacFawn et al. in HGSOC. In PFS > 1 year subgroup, the TLS-DSP signature exhibited a spatial distribution pattern nearly identical to Context 17, which displayed the highest TLS-DSP enrichment score among all contexts (**Figure 7F**, **Supplementary Figure 22A**). This spatial pattern was supported by multi-omic validation: GSEA confirmed significant post-treatment upregulation of the TLS-DSP program at the transcriptome level, and proteomic analysis demonstrated concordant increases in core TLS components, PTPRC, SELL, and PLEK, at the protein level (**Supplementary Figure 22B-C**). These data establish Context 17 as the spatial manifestation of TLS within the remodeled TME.

We next assessed the structural maturity of these TLS by mapping B cell differentiation gradients within Context 17 regions. In patient P17, we segmented distinct zones and analyzed expression gradients of lineage markers: MS4A1 (naïve/memory B cells), BCL6 (germinal center B cells), and JCHAIN/SDC1 (plasma cells). This revealed a clear follicular organization: Bn, Bm, and Bgc cell abundances decreased radially from core to periphery, while plasma cells increased markedly toward the outer edges (**Figure 7G**; **Supplementary Figure 22D**). This organized gradient from activated B cells to terminally differentiated plasma cells is a characteristic feature of functional TLS and was also observed in patient P14 (**Supplementary Figure 22E**).

Multiplex immunohistochemistry (mIHC) with CD3, CD8, CD20, and SDC1 provided independent histological validation (**Figure 7H**). Quantitative analysis demonstrated a

significant increase in morphologically identifiable TLS density in post-treatment lesions from patients with PFS >1 year versus their pre-treatment baselines, with a visible upward trend even in the PFS <1 year subgroup (**Figure 7H**). This protein-level confirmation using standard pathological criteria supports PASSION therapy-induced TLS neogenesis.

According to recent TLS classification, B:T-TLS, defined by dense, interactive B and T lymphocyte co-localization, represents the most functionally mature and prognostically relevant subtype (Cakmak et al., 2025). Context 17 exhibited the highest enrichment score for the B:T-TLS transcriptional signature and displayed a spatial pattern exclusive from PC-TLS-enriched regions (**Supplementary Figure 23**). Thus, Context 17 is not merely a B-cell-enriched area but a discrete spatial entity dedicated to active B-T cell collaboration, the spatial correlate of functional B:T-TLS in HGSOC.

To translate this discovery into a quantitative metric, we devised a Context 17 score. In the TCGA-OV cohort, a high score significantly correlated with improved overall survival and earlier disease stage (**Figure 7I**, **Supplementary Figure 24A-B, sTable 16**). This association was independently confirmed in the Birrer cohort (**Supplementary Figure 24C**). Within the PASSION trial, the Context 17 score was dynamically and specifically upregulated in the PFS >1 year subgroup (**Figure 7J**), establishing its utility as both a prognostic biomarker and an indicator of therapy-induced immune activation.

Collectively, these analyses, which span spatial context identification, structural maturation mapping, histological validation and multi-cohort prognostic verification, converge to identify Context 17 as the functional B:T-TLS correlate in HGSOC. Induction of this structure by the PASSION regimen thus constitutes a central immune mechanism, one in which therapeutic efficacy depends on the de novo assembly of a functionally competent B–T cell niche.

Enhanced Efficacy of PASSION plus PD-1 Blockade

Our multi-omic dissection demonstrated that PASSION therapy converts the "cold" ovarian TME into a "hot," TLS-enriched state. However, a subset of patients still experienced disease progression within one year, indicating that TLS formation alone does not guarantee durable tumor control. We therefore assessed the functional state of the therapy-induced immune response using the TIDE algorithm (Jiang et al., 2018). TIDE analysis confirmed robust immune activation post-PASSION: Merck 18 (IFN-γ) and CD8+ T cell signatures were significantly elevated, while MDSC activity was suppressed (**Supplementary Figure 25, sTable 17**). Strikingly, however, this immune infiltration was coupled with a marked increase in the T cell dysfunction score, a signature of exhausted, functionally impaired T cells, alongside rising CD274 (PD-L1) expression and composite TIDE scores. This countervailing profile indicates that the remodeled TME concurrently upregulates PD-1/PD-L1-mediated inhibitory pathways, potentially capping the anti-tumor efficacy of PASSION monotherapy.

These data provided a direct rationale for combining PASSION with immune checkpoint blockade. To quantify this potential, we applied the Enlight transcriptomic framework, a validated predictor of clinical response to PD-1/PD-L1 blockade (Dinstag et al., 2023; Hoang et al., 2024). Enlight scores were profoundly elevated in post-PASSION samples relative to pre-treatment baselines, whereas matched conventional NACT

samples showed a decline (**Figure 8A, sTable 18**), underscoring the unique immunomodulatory capacity of the PASSION regimen. Applying the established response threshold (Enlight score $\geqslant$ 0.54), we found that the proportion of predicted anti-PD-1 responders increased from 7.69% pre-treatment to 76.92% post-PASSION (p=0.001) (**Figure 8B**). These in silico results suggested that PASSION treatment may sensitize tumors to PD-1 blockade, supporting a sequential combination strategy of PASSION followed by anti-PD-1 (**Figure 8C**).

We validated this strategy in both BRCAmut and BRCAwt syngeneic HGSOC models. The PASSION plus anti-PD-1 combination significantly outperformed PASSION monotherapy in suppressing tumor growth and markedly reduced malignant ascites (**Figure 8D-H**). The enhancing effect of anti-PD-1 was even more pronounced in BRCAwt tumors (**Figure 8F, H**), a finding with substantial translational implications, as BRCAwt HGSOC is historically resistant to immunotherapy due to its profoundly immunosuppressive TME. Our data demonstrate that PASSION therapy can fundamentally reprogram this resistant state into an ICI-sensitive, immune-inflamed microenvironment, extending the promise of combination immunotherapy beyond the traditionally immunogenic BRCA-mutant subset.

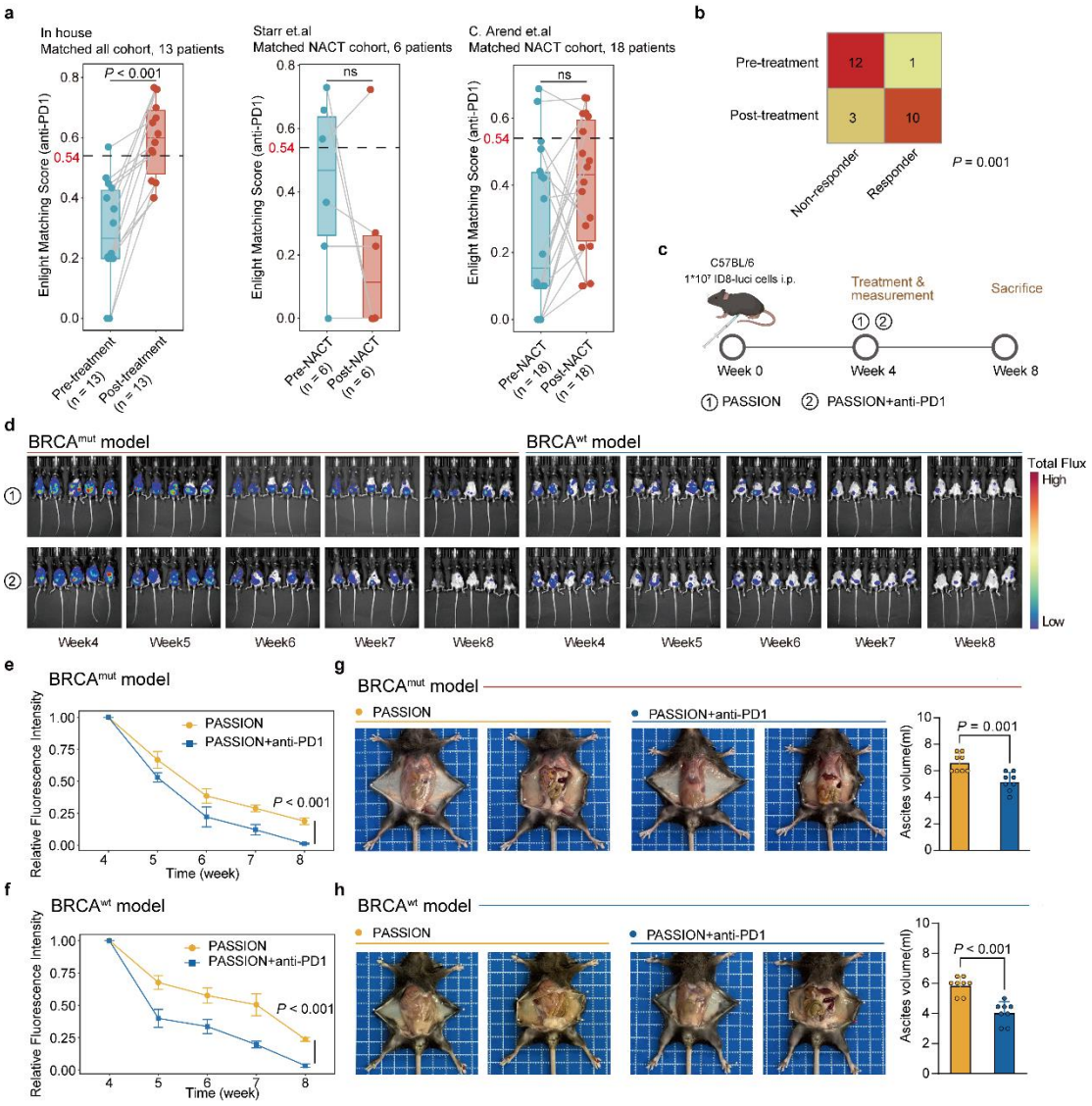


Figure 8 PASSION Therapy Sensitizes Ovarian Cancer to Anti-PD1 Blockade.

(**A**) Comparative analysis of anti-PD1 sensitivity scores pre- and post-treatment in the PASSION cohort (left) and NACT cohort (middle and right), calculated using the Enlight framework.

(**B**) Predicted distribution of anti-PD1-sensitive versus non-sensitive patients before and after treatment based on Enlight score stratification (threshold = 0.54).

(**C**) Schematic illustration of combination therapy with anti–PD1 in the mouse model.

(**D**) The bioluminescent images of all mice in each respective group.

(E, F) The temporal changes in BRCAmut (**E**) and BRCAwt (**F**) tumor burdens from week 4 to week 8 treatment period, as gauged by luminescence intensity, with treatment annotated by colors. Repeated-measures ANOVA followed by Bonferroni post-hoc multiple comparisons was used to evaluate longitudinal and intergroup differences.

(G, H) Ascites condition and anatomical diagrams of BRCAmut (**G**) and BRCAwt (**H**) model after treatment, with inter-group differences in ascites statistics analyzed by unpaired two-tailed Student's t-test on the right.

These results define a sequential therapeutic mechanism: (1) PASSION therapy remodels the TME and induces de novo TLS formation; (2) this immune activation triggers

compensatory PD-1/PD-L1 upregulation; (3) the addition of anti-PD-1 blockade releases this feedback inhibition, unlocking the full cytotoxic capacity of the TLS-educated anti-tumor immune response. Collectively, this provides a preclinical foundation for combining PASSION with ICIs and identifies a novel strategy to overcome immunotherapy resistance across the broad HGSOC population.

Discussion

Our study outlines a multi-step strategy to target the immunosuppressive TME of HGSOC, which contributes to its lethality and treatment resistance (Azzalini et al., 2023). This work, which integrated AI-augmented drug discovery, combination design, and mechanistic clinical translation, proposes a therapeutic strategy based on synergistic TME reprogramming. A central premise was that simultaneous disruption of the core signaling pathways in the HGSOC TME—CSF1R-driven myeloid suppression, VEGFR-driven pathological angiogenesis, and FGFR-mediated stromal and B cell signaling—would be required to disassemble this self-reinforcing, immunosuppressive network. To implement this multi-targeted strategy, we used an AI-driven screening platform, which identified surufatinib as an agent with concurrent activity against VEGFR, FGFR1, and CSF1R (Syed, 2021). We then combined surufatinib with the PARP inhibitor pamiparib to form the PASSION regimen. The combination was designed to act through two complementary mechanisms: surufatinib remodels the immunosuppressive stromal and vascular compartments, while pamiparib induces synthetic lethality in tumor cells and may engage innate immune pathways. The synergistic antitumor activity observed in preclinical models supported this combination strategy and provided a basis for clinical evaluation.

A key feature of this work was the use of the PASSION regimen in the neoadjuvant setting for advanced, initially unresectable disease. Unlike the conventional use of such targeted agents in later-line therapy, early neoadjuvant intervention produced high response and R0 resection rates, along with control of malignant ascites. This neoadjuvant approach suggests several advantages of frontline TME reprogramming. First, early vascular normalization and stromal remodeling can reduce tumor interstitial pressure, improve drug delivery, and alleviate physical barriers to surgical resection, helping convert unresectable disease into operable disease. Second, preoperative immune reprogramming converts immunosuppressive tumors into immune-inflamed lesions, thereby preserving host immune competence before surgical stress. Third, effective control of malignant ascites reduces perioperative morbidity, improves patient performance status, and enables safer, more timely interval debulking surgery. Fourth, deep and early tumor regression lowers tumor burden, eradicates micrometastases before they become established, and reduces the risk of intra-abdominal seeding during surgical manipulation. Fifth, induction of TLS neogenesis before surgery may establish durable anti-tumor immune memory that could synergize with subsequent adjuvant therapies and reduce early recurrence.

A key finding from the paired patient analysis was the therapy-induced formation of B:T-TLS after PASSION therapy (Cakmak et al., 2025). TLS assembly can be linked to several actions of the regimen: vascular normalization (via VEGFR inhibition) facilitates lymphocyte infiltration (Bottsford-Miller et al., 2012), myeloid reprogramming (via CSF1R

inhibition) alters chemokine production (Tong et al., 2024), and DNA damage response modulation (via PARP inhibition) may provide immunostimulatory signals (Li et al., 2026). These TLS, as sites of adaptive immune interaction, provide a spatial basis for the conversion of immunologically "cold" tumors into "hot," immune-inflamed lesions. A spatial signature (Context 17 score) derived from these TLS was associated with favorable survival in independent cohorts, suggesting its potential as a biomarker of productive immune remodeling.

Our data also show that PASSION-induced immune activation is accompanied by compensatory upregulation of checkpoint molecules such as PD-L1, indicating an adaptive resistance mechanism within the newly inflamed TME. This provided a rationale for the sequential addition of immune checkpoint blockade (Cabrita et al., 2020; Sautès-Fridman et al., 2019; Schumacher and Thommen, 2022). In preclinical models, the combination of PASSION with anti-PD-1 therapy improved antitumor efficacy, indicating that the regimen can prime the TME for enhanced response to immunotherapy. This effect was particularly evident in BRCA wild-type models, which are typically resistant to immunotherapy, suggesting that this approach can sensitize a broader patient population. These findings support a sequential strategy of TME reprogramming followed by checkpoint inhibition.

This study has limitations. The clinical cohort is small, although it was characterized in detail, and the specific molecular steps linking target inhibition to TLS assembly require further investigation. Larger, randomized trials with longer follow-up and dedicated studies on the kinome-wide effects of surufatinib will be needed to validate and refine this approach.

In conclusion, we report a translational approach that moves from the identification of core TME vulnerabilities to clinical application. By defining the PASSION regimen that reprograms the immunosuppressive TME, demonstrating its activity in preclinical models and patients, and identifying TLS neogenesis as a mechanistic correlate, we propose a treatment strategy for HGSOC. This approach of frontline, multi-targeted TME reprogramming followed by immunotherapy may improve outcomes for patients with this lethal malignancy.

Method

Cellline culture

The mouse ovarian cancer cell line ID8 was purchased from Sigma-Aldrich. ID8 cells were cultured in DMEM supplemented with 10% fetal bovine serum (FBS, Gibco) and penicillin-streptomycin solution (Thermo Fisher Scientific).

Cells were seeded into 12-well plates at a density of 1.3×10^5 cells per well. Negative control siRNA and BRCA1-targeting siRNA were obtained from Ribo Life Science. Tumor cells were transfected with these siRNAs using Lipofectamine 3000 (Thermo Fisher Scientific) following the manufacturer's protocol.

The siRNA sequences are listed below:

siBRCA1 #1 CCACACGATTTGACGGAAA #2 GCAGGAAATGGCTGAACTA

Animal Experiment

Six-to-eight-week-old female C57BL/6 mice were acclimatized before modeling. Ovarian cancer cells (80%–90% confluency) were digested, pelleted, washed twice with PBS, counted, and diluted to (2*10^7) cells/200 μL in PBS. Under head-down positioning, 200 μ

L cell suspension was injected intraperitoneally via a 1 mL sterile syringe at a 30°–45° oblique angle 0.5 cm beside the lower abdominal midline, with no fluid leakage after needle withdrawal.

At day 28 post-inoculation, abdominal hair from the sternal angle to the anus was depilated one day ahead of bioluminescence imaging. Mice were anesthetized with isoflurane, intraperitoneally injected with 200 μL 15 mg/mL D-luciferin potassium salt (DPBS formulation, stored cold and dark), and imaged supine 5–6 min later. Tumor engraftment was verified by bioluminescent signal intensity, and validated mice were randomly grouped for subsequent in vivo treatment assays.

Sulfatinib and pamiparib powders were obtained from Hutchison MediPharma (Shanghai) Co., Ltd. and BeiGene (Beijing) Co., Ltd., respectively.

0.5% sodium carboxymethyl cellulose (CMC-Na) vehicle was prepared 1 day ahead, aliquoted and stored at −20 °C. Sulfatinib was tested at 40 mg/kg in vivo: a 100 mg/mL DMSO stock was diluted 25-fold with CMC-Na, followed by 30 min sonication to make a 4 mg/mL working solution containing 4% DMSO, freshly prepared before dosing. Pamiparib was administered at 6.3 mg/kg: a 15.75 mg/mL DMSO stock was diluted 25-fold with vehicle and sonicated for 15–30 min at room temperature to generate a 0.63 mg/mL working solution with 4% DMSO, prepared on the day of treatment.

Experiment 1-Mice with validated intraperitoneal ovarian cancer xenografts were randomly allocated into four groups (n=10 per group) with the following oral gavage regimens administered twice daily for 28 consecutive days: (1) Vehicle control group: 4% DMSO diluted in 0.5% CMC-Na solution;(2) Pamiparib monotherapy group: 6.3 mg/kg pamiparib;(3) Sulfatinib monotherapy group: 40 mg/kg sulfatinib; (4) Combination group: 40 mg/kg sulfatinib plus 6.3 mg/kg pamiparib.

During the 28-day dosing period, general health status, food and water intake, hair condition and defecation were monitored daily. Body weight was recorded every day, abdominal circumference was measured once weekly, and in vivo bioluminescence imaging was performed once per week to monitor tumor burden.

Experiment 2-Tumor-bearing mice received the following treatments for 28 days (n=10 per group): (1) Dual-agent combination group: twice-daily oral gavage of 40 mg/kg sulfatinib combined with 6.3 mg/kg pamiparib; (2) Triple combination group (sulfatinib + pamiparib + anti-PD-1): The same oral dosing of sulfatinib and pamiparib as group (1), plus intraperitoneal injection of 200 μg anti-PD-1 antibody on day 7, day 14, day 21 and day 28.

Consistent monitoring was performed as described in Experiment 1, including daily body weight recording, weekly abdominal circumference measurement and weekly bioluminescence imaging, with continuous observation of mouse physical signs throughout treatment.

All animals were sacrificed after 28 days of continuous treatment. A 0.5% sodium pentobarbital solution was prepared in advance, and mice were anesthetized via intraperitoneal injection at a dose of 50 mg/kg sodium pentobarbital. Once fully anesthetized, each mouse was secured on a dissection platform. The abdominal skin was dissected while preserving the intact peritoneum for photographic documentation. Abdominal circumference and abdominal sagittal height (vertical distance from the highest abdominal prominence to the horizontal plane in supine position) were measured and

recorded.

Blood samples were collected via orbital exsanguination. The peritoneal membrane was incised, and ascites fluid was aspirated into centrifuge tubes using a syringe. Visible tumor nodules were counted, photographed and weighed. Fresh tumor fragments were transferred to cryovials, wrapped in aluminum foil, snap-frozen in liquid nitrogen, and stored long-term at −80 °C.

Subsequently, the thoracic cavity was opened to expose the heart for transcardial perfusion: the circulatory system was first perfused with pre-cooled PBS, followed by 4% paraformaldehyde (PFA) for tissue fixation. Peritoneal tumor tissues, heart, liver, spleen, lung, kidney and ovary were harvested and fixed in 4% PFA for subsequent paraffin embedding, sectioning and histological staining.

Study design and paticipants

PASSION is a prospective single-centre, single-arm, phase 2 trial done in the First Affiliated Hospital of University of Science and Technology of China (USTC) in China. Enrolled patients were aged at least 18 years or older; had newly diagnosed, histologically confirmed advanced and unresectable cancer of the ovary, peritoneum, or fallopian tube (collectively defined as ovarian cancer); were classified as stage III or IV according to the criteria of the International Federation of Gynecology and Obstetrics (FIGO 2018); were ineligible for R0 tumor debulk surgery (defined as meeting at least one of following criteria: Fagotti endoscopic score⩾8, Upper abdomen CT score⩾3,) nor tolerable surgery (defined as meeting at least one of following criteria: age ⩾ 80, body mass index [BMI] ⩾ 40.0, multiple chronic diseases, hypoproteinemia, moderate to massive ascites, newly diagnosed venous thromboembolism, or Eastern Cooperative Oncology Group [ECOG] performance status 2); had measurable disease according to Response Evaluation Criteria in Solid Tumors (RECIST, version 1.1); had an ECOG performance status of 0-2; had adequate hematological (absolute neutrophil count, platelets, and hemoglobin), renal (creatinine or creatinine clearance), and liver (total bilirubin, aspartate aminotransferase, and alanine aminotransferase) function; had a life expectancy of more than 12 weeks.

Exclusion criteria included previous exposure to PARP inhibitors or known allergies to any of the excipients, a major surgical procedure within 3 weeks before day 1 of this study, previous malignancy (except adequately treated basal cell, squamous cell skin cancer, superficial or in-situ cancer of breast and cervix) within 3 years before day 1 of this study. Patients with one or more of the following conditions were also excluded: significant cardiovascular disease, unstable angina, relevant gastrointestinal disease, or a history of HIV.

This study was approved by Ethics Committee of Anhui Cancer Hospital and in accordance with the Declaration of Helsinki. All patients provided written informed consent before enrolment. The trial is registered with ClinicalTrials.gov, NCT05652283.

Procedures

This study consisted of three cycles of neoadjuvant therapy followed by surgical removal if feasible followed by adjuvant chemotherapy. Patients who had progressive disease with NAT received additional chemotherapy.

NAT was given as follows: pamiparib 40 mg twice daily oral administration on days 1-63 in cycles A1-A3; surufatinib 250 mg once daily oral administration on days 1-42 in cycles A1-A2.Chemotherapy was given as follows: carboplatin, intravenous infusion in 24h on days 1, 22 and 43 in cycle B, on days 1, 22, 43 and 64 in cycle C; paclitaxel, intravenous infusion in 24h on days 1, 22 and 43 in cycle B, on days 1, 22, 43 and 64 in cycle C. Dose adjustments at the NAT phase were based on haematological and non-haematological toxic effects. In any event of grade 3 or 4, treatment was delayed along with active management until resolution to grade 1 or 2.

Patients were monitored every week in cycles A1-A3, and every three weeks during the chemotherapy period, which consisted of evaluations of performance status, adverse events, and laboratory procedures. Adverse events were monitored throughout the treatment period and were graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE, version 5.0). Laboratory procedures (haematology, chemistry, urinalysis) were done on days 0, 7, 14, 21, 28, 35, 42, 49, 56 and 63 in cycles A1-A3, on days 1, 22 and 43 in cycle B, on days 1, 22, 43 and 64 in cycle C.

Patients were required to provide a core or excisional tumour biopsy (before the initiation of NAT [pre-treatment sample] and at the time of definitive surgery [post-treatment sample]) and blood sample at the time of laboratory procedures for explosive analyses. Pre-treatment and resected tumor samples were chosen for bulk, single-nuclear and spatial RNA sequencing and mass spectrometry analysis of protein.

Outcomes

The primary outcome measure was R0 resection margin rate. R0 resection was measured based on the surgery medical record.

The secondary outcome measures were objective response rate (the proportion of patients who had either complete or partial response according to RECIST criteria), the rate of pathological complete response, 1-year progression-free survival (time from first treatment to progression or death), 1-year overall survival (time from first treatment to death due to any cause), adverse events to assess the safety and side-effect profile.

Bulk RNAseq

RNA Extraction, library construction and sequencing

Total RNA was extracted using the Trizol reagent kit (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's protocol. RNA quality was assessed with an Agilent 2100 Bioanalyzer (Agilent Technologies, Palo Alto, CA, USA) and confirmed via RNase-free agarose gel electrophoresis. Following extraction, eukaryotic mRNA was enriched using Oligo(dT) beads. The enriched mRNA was then fragmented using fragmentation buffer and reverse transcribed into cDNA with the NEBNext Ultra RNA Library Prep Kit for Illumina (NEB #7530, New England Biolabs, Ipswich, MA, USA). The resulting double-stranded cDNA fragments were purified, end-repaired, adenylated, and ligated to Illumina sequencing adapters. The ligation reaction was purified with AMPure XP Beads (1.0X). Ligated fragments were size-selected by agarose gel electrophoresis, PCR amplified, and sequenced using the Illumina NovaSeq 6000 platform by Gene Denovo Biotechnology Co. (Guangzhou, China).

Quantification of gene abundance

Reads obtained from the sequencing machines includes raw reads containing adapters or low-quality bases which will affect the following assembly and analysis. Thus, to get high quality clean reads, reads were further filtered by fastp (version 0.18.0).2 Short reads alignment tool Bowtie2 (version 2.2.8) was used for mapping reads to ribosome RNA (rRNA) database.3 The rRNA mapped reads then will be removed. The remaining clean reads were further used in assembly and gene abundance calculation. An index of the reference genome was built, and paired-end clean reads were mapped to the reference genome using HISAT2. 2.4 and other parameters set as a default.4 The mapped reads of each sample were assembled by using StringTie v1.3.1 in a reference-based approach.5,6 For each transcription region, a FPKM (fragment per kilobase of transcript per million mapped reads) value was calculated to quantify its expression abundance and variations, using RSEM software.

Data analysis

TIDE algorithm analysis was performed on bulk transcriptomic profiles of pre- and post-neoadjuvant treatment samples (http://tide.dfci.harvard.edu/)(Jiang et al., 2018). Gene expression data were log2-transformed and standardized according to the official TIDE pipeline. Core signatures including T cell dysfunction, T cell exclusion, CD8+ T cell infiltration, MDSC abundance, IFN-gamma (Merck18) signature, and PD-L1 (CD274) expression were quantified to characterize immune status and immune-evasion phenotypes.

To quantitatively evaluate the sensitivity of tumor samples to immune checkpoint blockade, the ENLIGHT Matching Score was computed using the validated ENLIGHT transcriptome-based prediction framework (https://ems.pangeabiomed.com/)(Dinstag et al., 2023). Bulk RNA-sequencing expression matrices were log2-transformed, quantile-normalized, and formatted to match the model input requirements. The ENLIGHT model integrates gene interaction networks, immune-related signatures, and target expression levels to generate a continuous matching score, where higher values indicate greater predicted clinical benefit from therapy. For clinical stratification, the predefined threshold of score $\geqslant$ 0.54 was used to classify samples as ICB-responsive candidates.

Mass spectrometry analysis of protein

Protein extraction

Appropriate samples were suspended in lysis buffer (1% sodium deoxycholate, 8M urea) containing a 1x protease inhibitor cocktail to inhibit protease activity. The mixture was vortexed thoroughly and treated twice with a high-throughput tissue grinder. The samples were then incubated at 4°C for 30 minutes, with vortexing every 10 minutes. Following centrifugation at 16,000g for 20 minutes at 4°C, the protein concentration in the supernatant was determined using the bicinchoninic acid (BCA) method with a BCA Protein Assay Kit, following the manufacturer's protocol.

Zymolysis

Sample preparation involved protein denaturation, reduction, alkylation, tryptic digestion,

and peptide cleanup. A commercially available iST Sample Preparation kit (PreOmics, Germany) was used following the manufacturer's protocols. Briefly, 50 µL of lysis buffer was added to the sample and heated at 95°C for 10 minutes at 1000 rpm with agitation. After cooling to room temperature, trypsin digestion buffer was added, and the sample was incubated at 37°C for 2 hours at 500 rpm with shaking. The digestion process was stopped with stop buffer. Sample cleanup and desalting were performed using the iST cartridge with the recommended wash buffers. Peptides were eluted with elution buffer (2 × 100 µL) and then lyophilized using a SpeedVac.

Data acquisition

The UltiMate 3000 liquid chromatography system (Thermo Fisher Scientific, MA, USA) was connected to the timsTOF Pro 2, an ion-mobility spectrometry quadrupole time-of-flight mass spectrometer (Bruker Daltonics). Samples were reconstituted in 0.1% formic acid (FA), and 200 ng of peptide was separated using an AUR3-15075C18 column (15 cm length, 75 μm i.d., 1.7 μm particle size, 120 Å pore size, IonOpticks) with a 60-minute gradient starting at 4% buffer B (80% acetonitrile with 0.1% FA). The gradient increased stepwise to 28% over 25 minutes, 44% over the next 10 minutes, and 90% over another 10 minutes, held for 7 minutes, then equilibrated at 4% for 8 minutes. The column flow rate was maintained at 400 nL/min with a column temperature of 50°C.

Data-independent acquisition (DIA) data were acquired in the diaPASEF mode. We defined 22 × 40 Th precursor isolation windows ranging from m/z 349 to 1229. To adapt the MS1 cycle time, we set the repetitions to variable steps in the 13-scan diaPASEF scheme. During PASEF MS/MS scanning, the collision energy was ramped linearly as a function of the mobility from 59 eV at 1/K0 = 1.6 Vs/cm^2 to 20 eV at 1/K0 = 0.6 Vs/cm^2.

Data Analysis

Data from DIA were processed and analyzed using Spectronaut 18 (Biognosys AG, Switzerland) with default settings. The database used was Uniprot. Specific trypsin was designated as the digestion type and enzyme. Carbamidomethylation of cysteine was set as the fixed modification, and oxidation of methionine as the variable modification. Retention time prediction was set to dynamic iRT. Data extraction was performed by Spectronaut based on extensive mass calibration, dynamically determining the ideal extraction window according to iRT calibration and gradient stability. The q-value (FDR) cutoff was set at 1% for both precursor and protein levels. Decoy generation was configured to 'mutated,' which is similar to scrambled but applies a random number of amino acid position swaps (minimum=2, maximum=length/2). The normalization strategy was set to local normalization. Peptides passing the 1% q-value cutoff were used to calculate major group quantities using the MaxLFQ method.

Single-cell RNA-seq data processing

snRNA-seq data were processed using Cell Ranger (10x Genomics) with the GRCh38 human reference genome for raw data alignment, barcode processing, and gene expression matrix generation. Subsequent quality control and downstream analysis were performed in the Seurat R package (v4). Cells were retained if they met the following criteria: >200 detected genes, 500–30,000 unique molecular identifiers, and <15% mitochondrial gene content to exclude low-quality cells and apoptotic events. Data were

normalized, scaled, and integrated across samples using the standard Seurat integration pipeline to minimize technical batch effects. Unsupervised dimensionality reduction was conducted using principal component analysis, followed by UMAP visualization and graph-based clustering to identify distinct cell populations. Major cell lineages were annotated based on cluster-specific expression of canonical marker genes. For targeted subset analysis (e.g., endothelial cells, B cells), cells of interest were subset, then re-analyzed using identical PCA, dimensionality reduction, and visualization parameters. Cluster-defining marker genes were identified using the FindAllMarkers function in Seurat with default settings (logFC threshold = 0.25, adjusted $P < 0.05$).
Subclusters of endothelial cells and B cells were further resolved by computing gene signature enrichment scores using the AddModuleScore function in Seurat, based on established lineage- and function-specific gene signatures from published studies.

Cell–cell communication analysis

Cell–cell communication networks were inferred using CellChat(Jin et al., 2021). Normalized single-cell expression matrices were imported into CellChat, and the standard analysis workflow was executed with default parameters. Ligand–receptor pairs and signaling pathways were annotated using the built-in CellChat database. The netAnalysis_contribution function was used to quantify the contribution of individual ligand–receptor pairs to overall signaling. The netAnalysis_signallingRole function was applied to define sender and receiver roles for each cell population. Inferred communication networks were visualized using the netVisual_aggregate function for integrated pathway-level communication patterns.

Spatial transcriptomics data collection and preprocessing

fresh-frozen clinical specimens were analyzed using the 10x Genomics Visium spatial transcriptomics platform to capture transcriptome profiles while preserving tissue spatial architecture. Briefly, 2–3 tumor tissue pieces were embedded cut-side down in optimal cutting temperature (OCT) compound in plastic molds. The OCT-embedded blocks were immediately snap-frozen in pre-chilled isopentane and stored at −80 °C until sectioning. Cryosections were cut at 10 μm thickness and processed for H&E staining, tissue permeabilization, and spatial library construction. Tissue permeabilization was optimized at 10 minutes; all tissue optimization and library preparation steps were performed strictly following the manufacturer's protocols. Barcoded spatial libraries were sequenced on an Illumina NovaSeq 6000 platform using PE150 mode. Raw sequencing data were processed for quality control, aligned, and quantified using Space Ranger (v1.1) with the human reference genome.

Spatial deconvolution using CARD

To further define the spatial localization of single-cell clusters within TME, we performed cell-type deconvolution of spatial transcriptomic profiles using the CARD R package, guided by single-cell transcriptomic atlas (Tanevski et al., 2022). Briefly, a CARD object was first constructed using the CreateCARDObject function, with single-cell data serving as the reference signature matrix. Deconvolution was then executed using the

CARD_deconvolution function under default parameters to quantitatively infer the proportional abundance of each cell cluster within every spatial spot.

Spatial transcriptomic data analysis

Spatial transcriptomic gene-by-spot matrices were analyzed using the Seurat R package (v4). Prior to downstream analysis, no spots were excluded. Normalization across spatial spots was performed using the SCTransform function.

For each spatial transcriptomic sample, SPATA2 was used to distinguish between tumor and non-tumor regions based on spatial location and gene expression characteristics(Kueckelhaus et al., 2024). Only non-tumor spots were retained for subsequent spatial context analysis. To characterize distinct cellular spatial contexts, PCA was first performed using the RunPCA function, followed by dimensionality reduction via UMAP with the RunUMAP function; both steps were based on cell-type proportions inferred by CARD. A total of 17 unique spatial contexts were identified using the FindNeighbors and FindClusters functions with a clustering resolution of 1.0.

Spatial gene set scoring was performed using the AddModuleScore function in the Seurat package. For the Context 17 score, signature genes were defined by separately running the FindAllMarkers function on spatial spots assigned to Context 17 across all samples, followed by intersecting the top 100 marker genes from each sample to obtain a robust consensus signature.

Survival analysis using the TCGA-OV cohort

Gene expression quantification (fragments per kilobase of transcript per million mapped reads, FPKM) and corresponding clinical annotations for ovarian cancer cohort (TCGA-OV) were retrieved from the UCSC Xena platform (http://xena.ucsc.edu/). Patients were stratified into high-score and low-score subgroups using the median signature score as the cutoff. Kaplan–Meier survival curves were plotted using the survival R package (v3.2.11), and differences in overall survival between groups were evaluated by the log-rank test. A P value < 0.05 was considered statistically significant.

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Author Contributions

W.X., B.X., F.W. and W.P. conceived and funded this project. B.X., W.J., Y.C., W.P., W.S., Y.T., W.W. and B.H. conducted patient management, clinical data sorting and sample collection for the clinical trial. Y.C., Q.Y., X.H., W.S., Y.S., H.S., H.M., H.Z., X.L., Y.W., T.H. and M.T. collected clinical samples and carried out animal experiments. W.P. and W.S. performed data analysis. W.X. and W.P. drafted the manuscript.

Resource Availability

Lead Contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Wei Xiong (wxiong@ustc.edu.cn).

Materials Availability

All unique/stable reagents generated in this study are available from the lead contact with a completed materials transfer agreement.

Data and Code Availability

De-identified individual participant data generated during this PASSION trial, including clinical efficacy, safety, multi-omics sequencing, and biomarker datasets, will be available to qualified researchers upon reasonable written request submitted to the corresponding author.

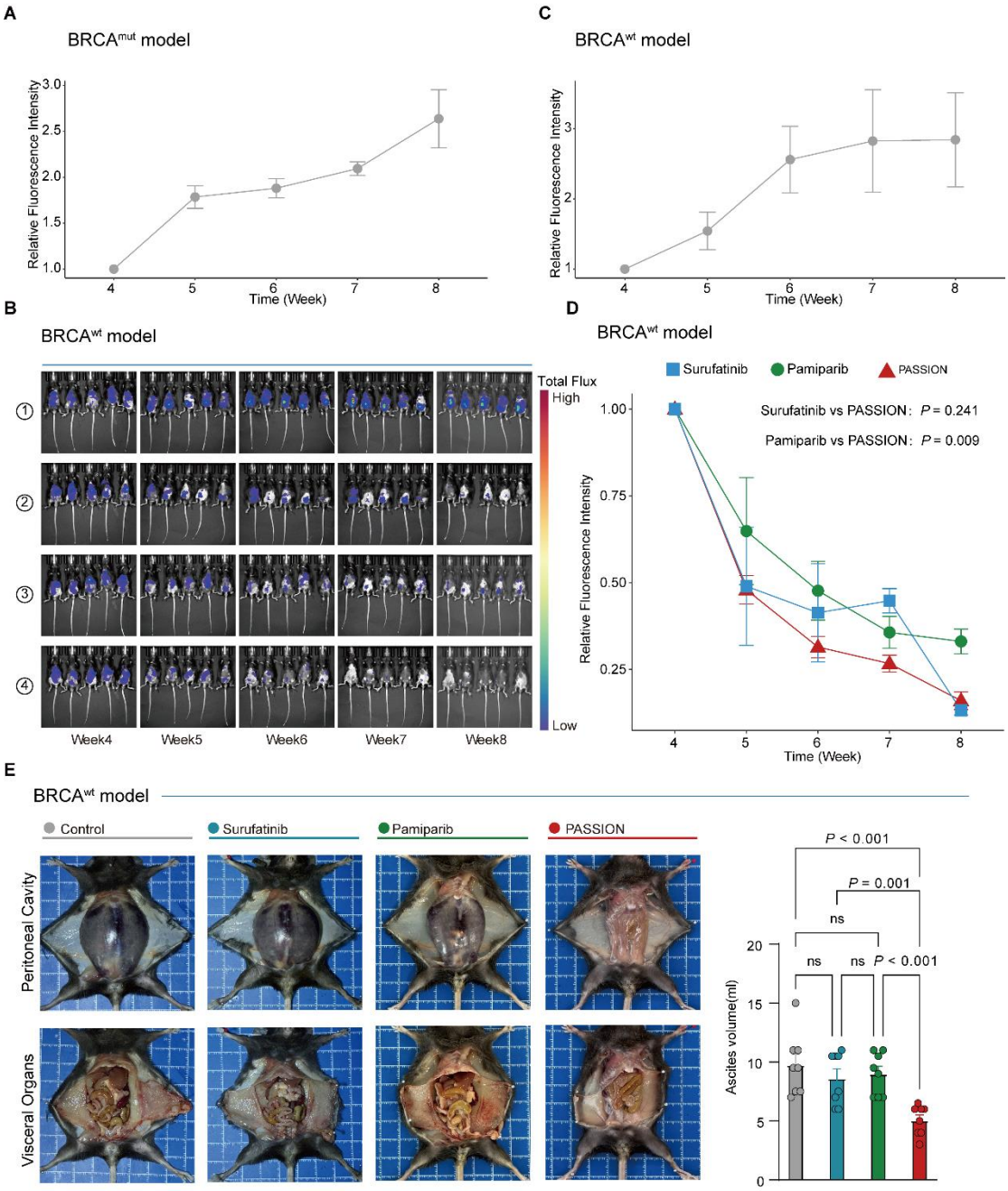
Supplemental figure

Click here to access/download;Supplemental figure;202607sFigure.docx

Supplementary Information

Multi-target TME Reprogramming Awakens TLS driven Immunity in Ovarian Cancer

Wenju Peng, Yao Chen, Wenjing Jiang, Qiongli Yu, Xu Huang, Wulin Shan, Wei Shen, Yuan Tian, Yuxuan Shi, Hanxue Sun, Huanhuan Mao, Haomin Zhang, Xiaofei Lu, Yumeng Wang, Wenjun Wu, Baoli Hong, Ting Han, Mengxi Tang, Feng Wu, Bairong Xia & Wei Xiong



Supplementary Figure 1. Therapeutic Effects of Combination Treatment in BRCA-mutant and BRCA-wild-type Ovarian Cancer Mouse Models, related to Figure 2

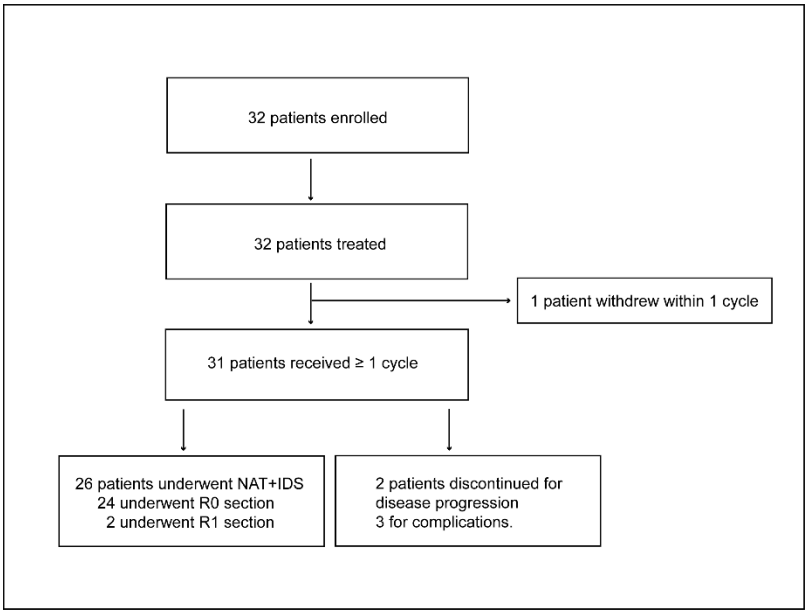
(A) The temporal changes in BRCAmut tumor burdens from week 4 to week 8 treatment period of control group.

(B) The bioluminescent images of all mice in each respective group.

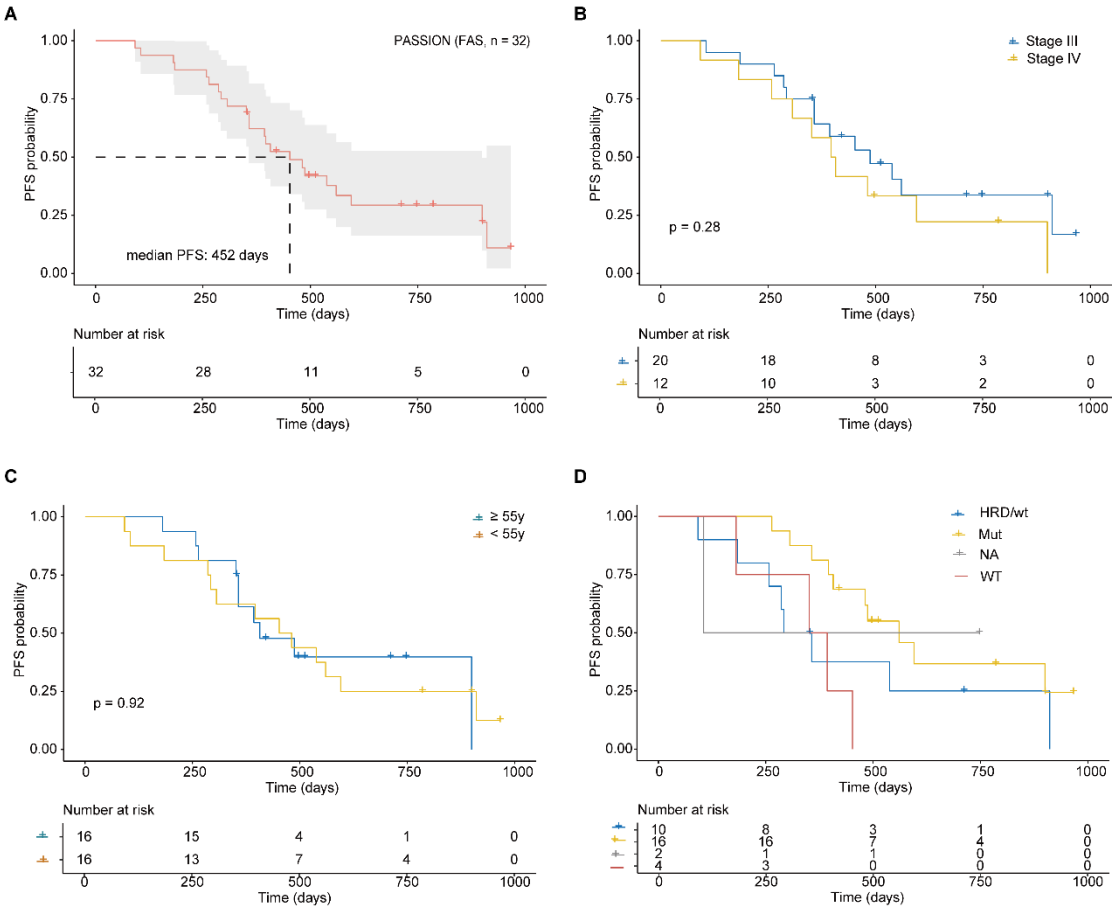
(C) The temporal changes in BRCAwt tumor burdens from week 4 to week 8 treatment period of control group.

(D) The temporal changes in BRCAwt tumor burdens from week 4 to week 8 treatment period, as gauged by luminescence intensity, with treatment annotated by colors.

(E) Ascites condition and anatomical diagrams of BRCAwt (left) model after treatment, with inter-group differences in ascites statistics (right).



Supplementary Figure 2. PASSION diagram, related to Figure 3

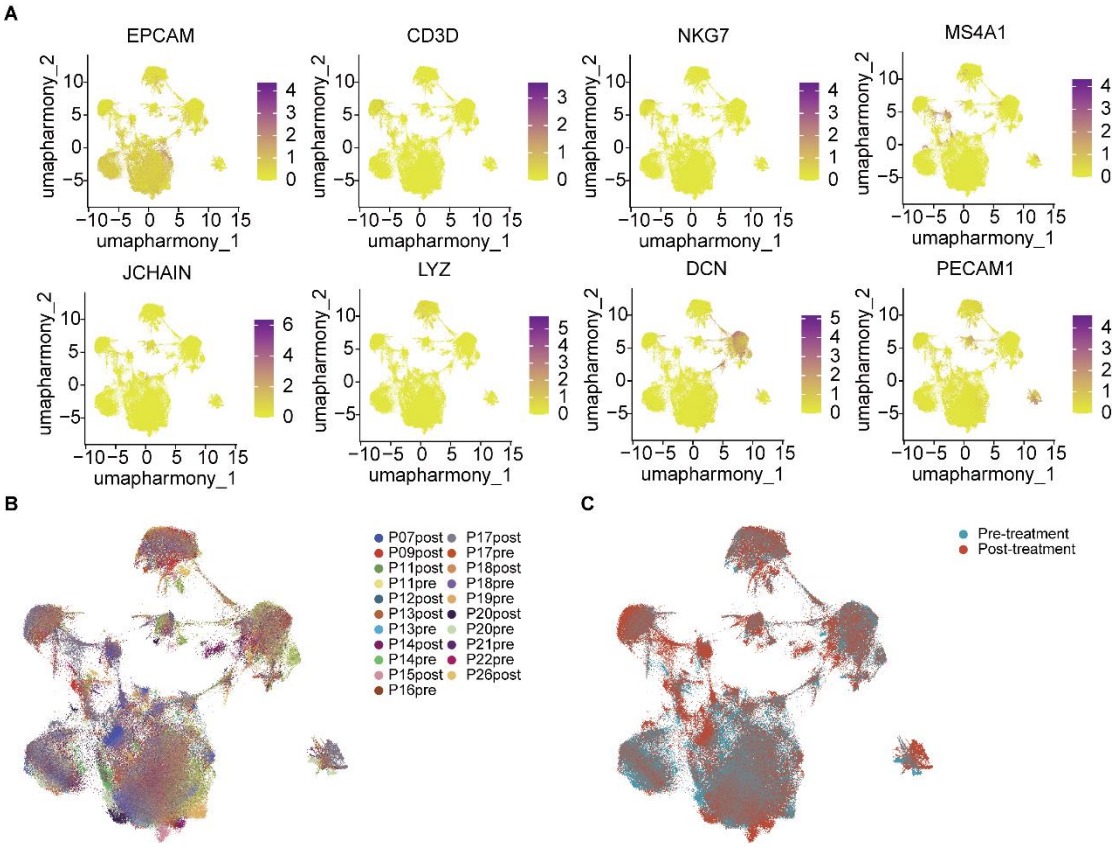


Supplementary Figure 3. Kaplan-Meier analyses of PFS in the PASSION cohort, related to Figure 3

(A) Kaplan-Meier curve for PFS in the PASSION cohort (FAS, n=32). The median PFS was 429.5 days.

(B-D) Kaplan-Meier curves for PFS in key patient subgroups. Subgroup analyses showed

no significant difference in PFS between patients with stage III and stage IV disease (B) or by age (C). BRCA mutation status remained a prognostic factor (D). P values were calculated using the log-rank test.

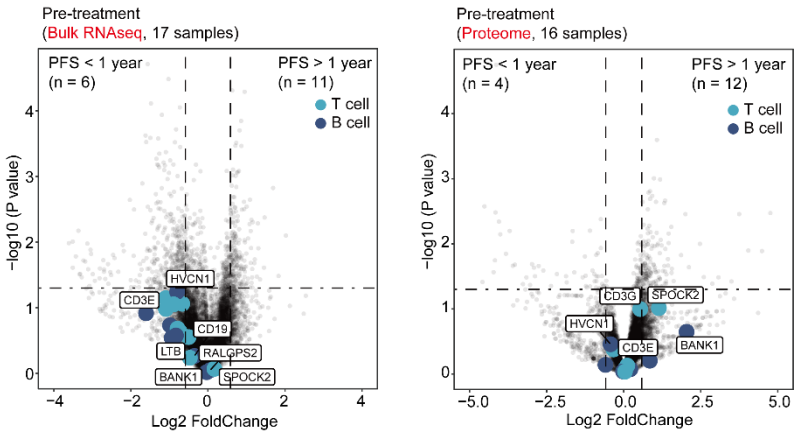


Supplementary Figure 4. Dimensional Reduction UMAP plots of snRNA-seq Profiles stratified by Cell Subtype, Sample and Sampling Time, related to Figure 4

(A) UMAP visualization showing the expression of canonical marker genes across major cell type.

(B) UMAP visualization of snRNA-seq data, with sample annotated by distinct colors.

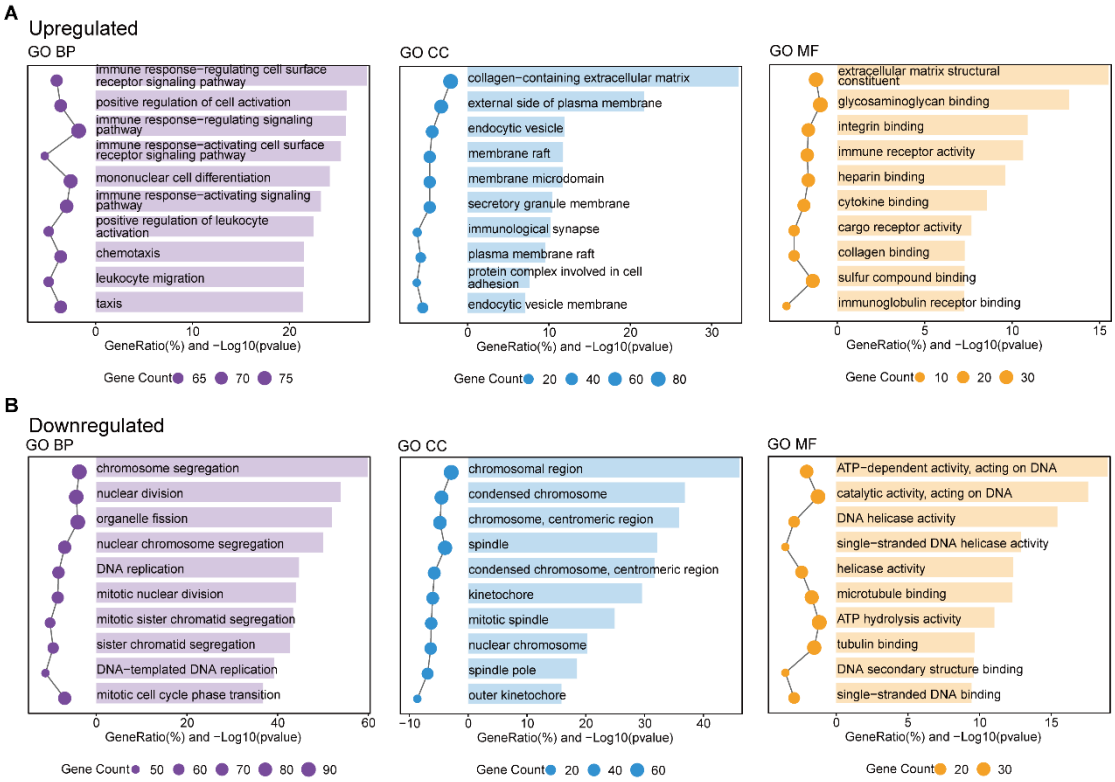
(C) UMAP visualization of snRNA-seq data, with sample annotated by sampling time.



Supplementary Figure 5. PFS-stratified Pre-treatment Multi-omic Differential

Expression Profiles, related to Figure 4

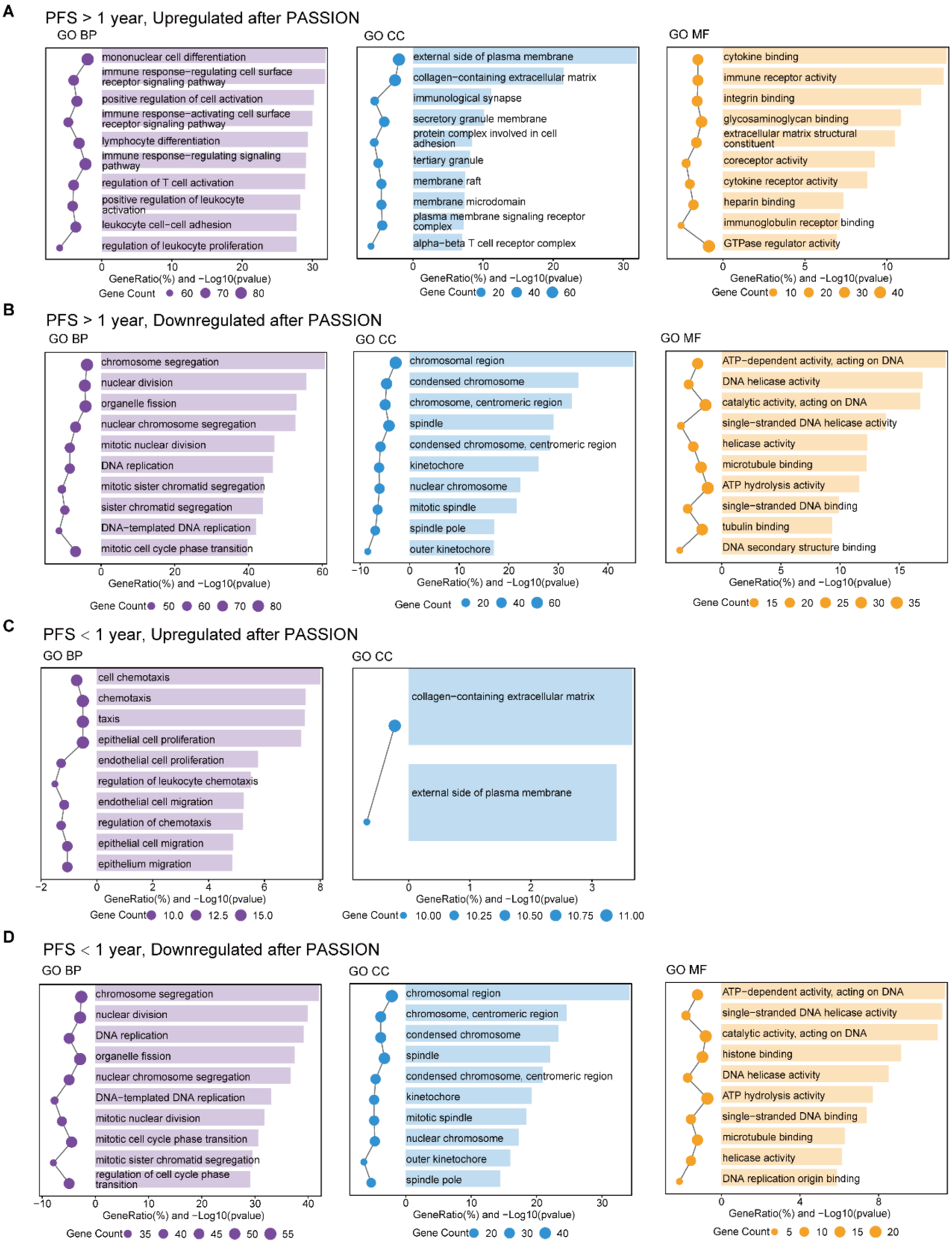
Volcano plot of proteomic data showing differentially expressed genes (left) and proteins (right) between pre-treatment samples in sample-matched patients of PFS > 1 year and PFS < 1 year subgroups, with T cell– and B cell–related proteins annotated by color.



Supplementary Figure 6. GO Enrichment Analysis , related to Figure 4

(A) Bar plot shows enriched pathways from GO BP (left), CC (middle) and MF (right) using upregulated DEGs. P values are derived from clusterProfiler package.

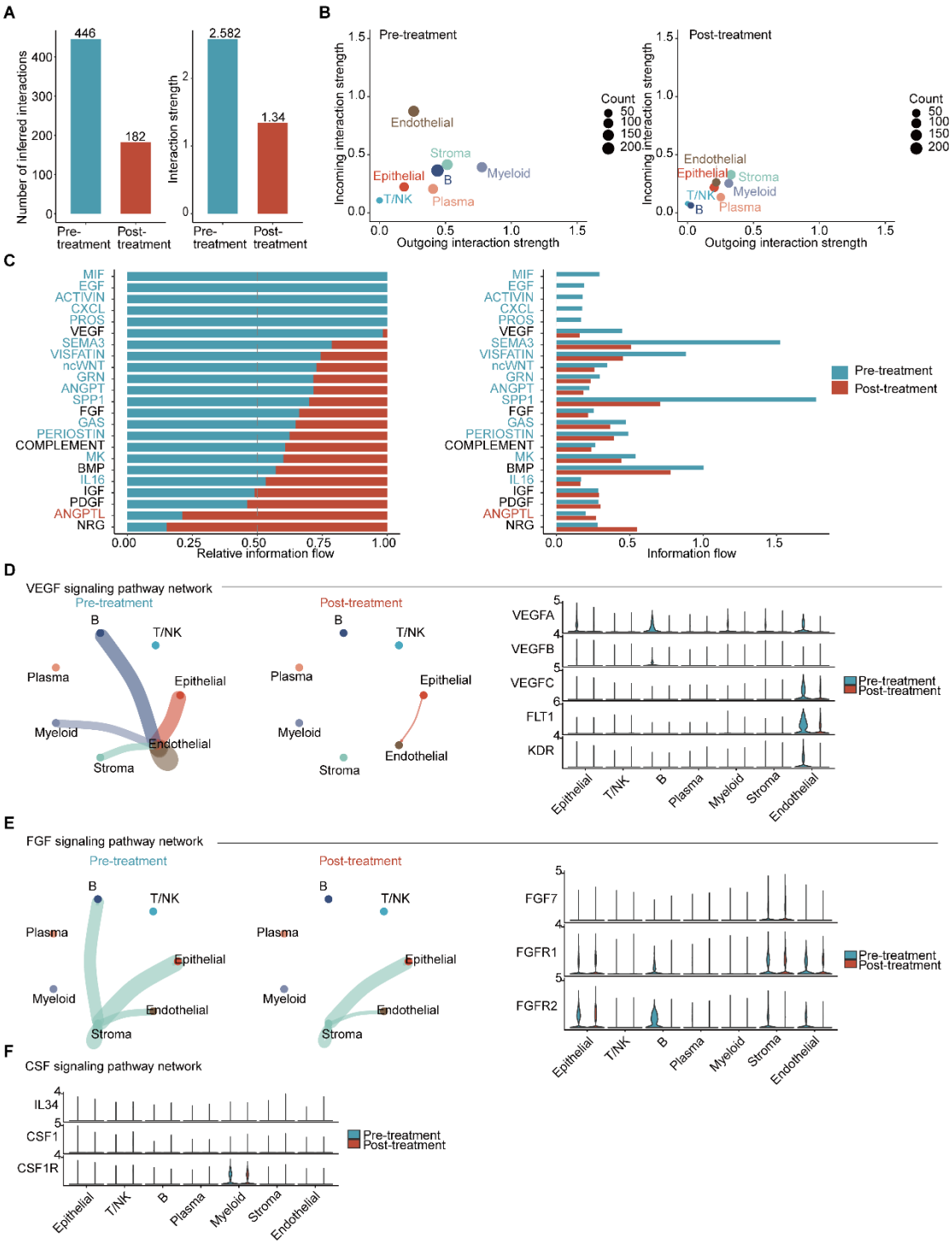
(B) Bar plot shows enriched pathways from GO BP (left), CC (middle) and MF (right) using downregulated DEGs. P values are derived from clusterProfiler package.



Supplementary Figure 7. GO Enrichment Analysis with PFS, related to Figure 4

(A-B) Bar plot shows enriched pathways from GO BP (left), CC (middle) and MF (right) using upregulated (A) and downregulated (B) DEGs in patients with PFS > 1 year. P values are derived from clusterProfiler package.

(C-D) Bar plot shows enriched pathways from GO BP (left), CC (middle) and MF (right) using upregulated (C) and downregulated (D) DEGs in patients with PFS < 1 year. P values are derived from clusterProfiler package.



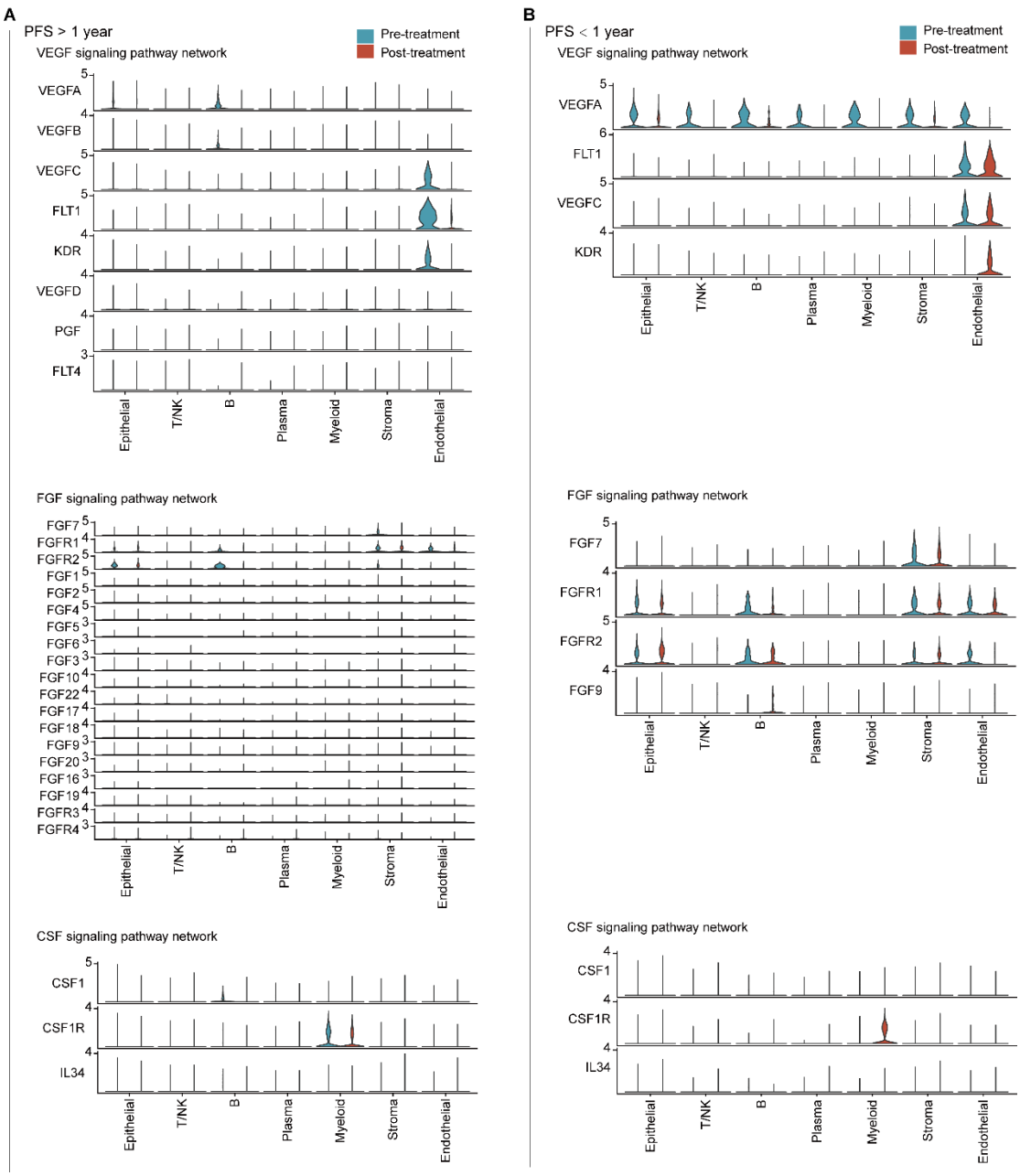
Supplementary Figure 8. Cell-cell Communication Analysis, related to Figure 4

(A) Bar plot showing total number of interactions (left) and interaction strength (right) of the inferred cell-cell communication networks from pre- (blue) and post-treatment (red) samples.

(B) Dot plot showing outgoing and incoming interaction strength pattern of the major cell populations in pre- (left) and post-treatment (right) samples.

(C) Comparative analysis of enriched ligand pathways, bar plot showing the relative (left) and enriched ligand-receptor pathways (right) between pre- and post-treatment samples. Significant signaling pathways were ranked based on differences in the overall

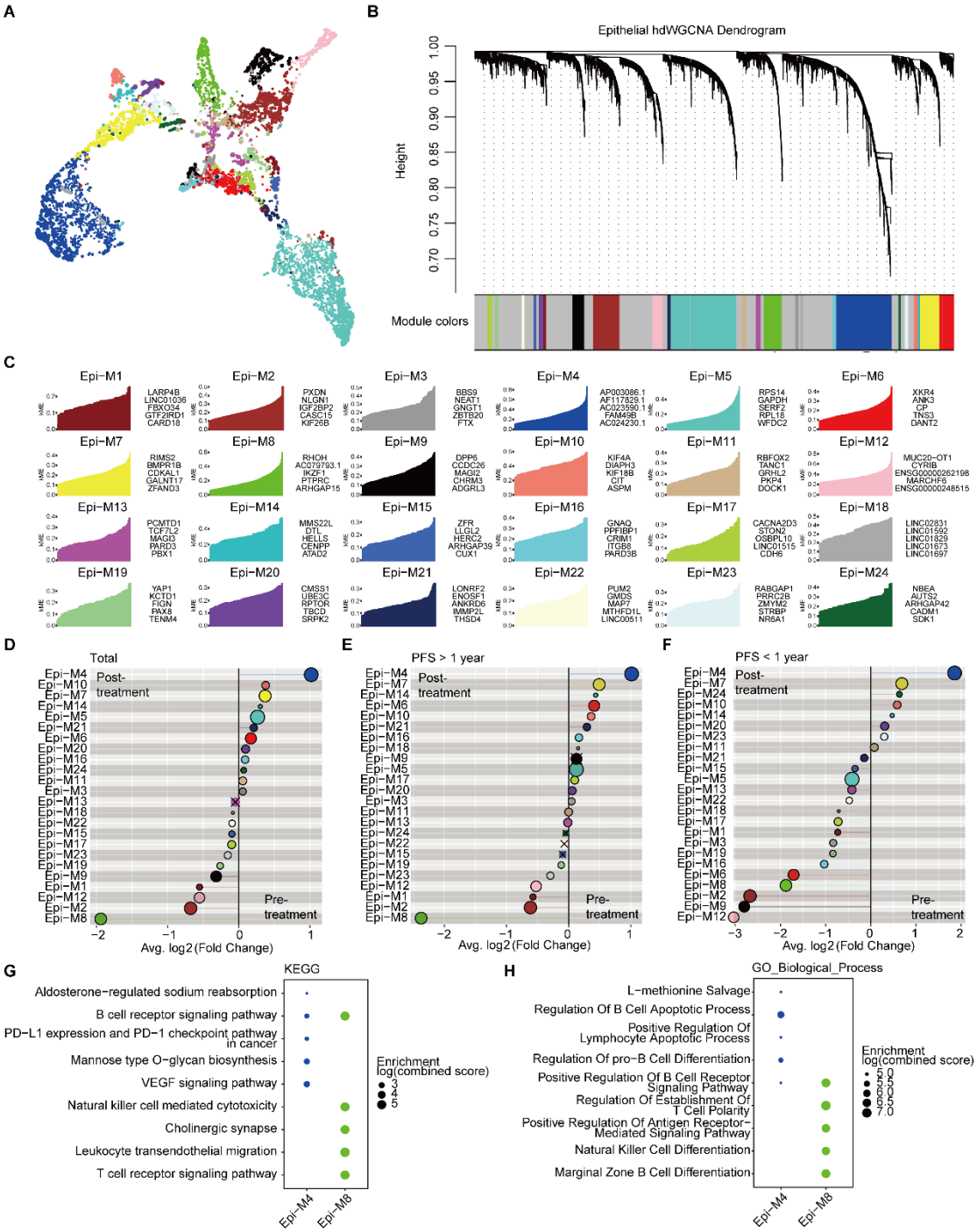
information flow within the inferred networks between pre- (blue) and post-treatment (red) samples. P values were calculated with paired Wilcoxon test.
(D) Circle plot illustrating interaction pattern of VEGF signaling pathway in pre- (left) and post-treatment (right) samples. Violin plot showing expression levels of ligands and receptors of VEGF signaling pathway (right panel).
(E) Circle plot illustrating interaction pattern of FGF signaling pathway in pre- (left) and post-treatment (right) samples. Violin plot showing expression levels of ligands and receptors of FGF signaling pathway (right panel).
(F) Violin plot showing expression levels of ligands and receptors of CSF signaling pathway.



Supplementary Figure 9. Expression levels of Ligands and Receptors in Targeted Pathways, related to Figure 4

(A) Violin plot showing expression levels of ligands and receptors of VEGF, FGF and CSF signaling pathway in patients with PFS > 1 year.

(B) Violin plot showing expression levels of ligands and receptors of VEGF, FGF and CSF signaling pathway in patients with PFS < 1 year.



Supplementary Figure 10. hdWGCNA Analysis of Epithelial Compartment, related to Figure 4

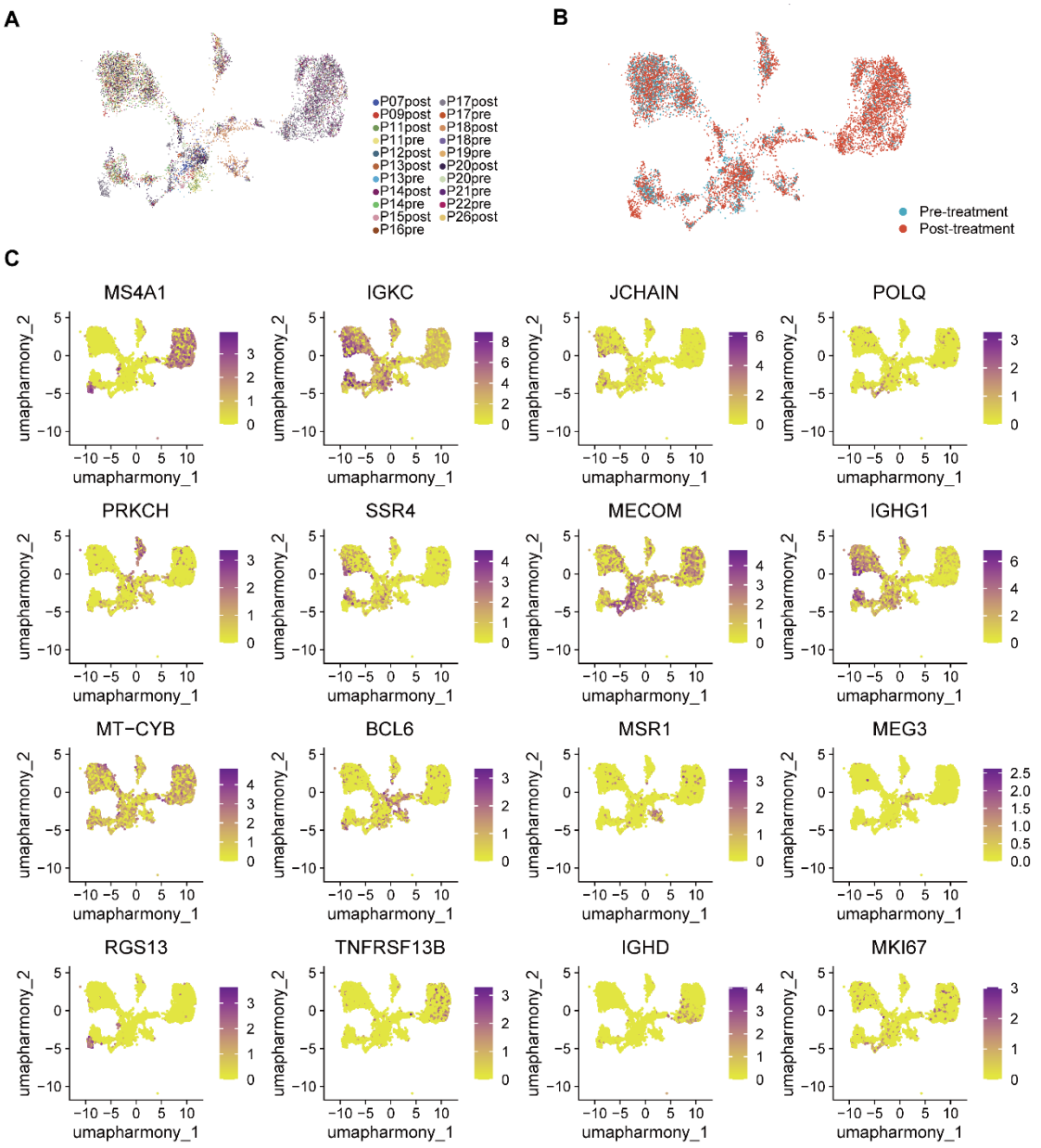
(A) UMAP visualization showing the genes and their co-expression relationships, colored by co-expression modules.

(B) WGCNA dendrogram showing the different co-expression modules resulting from the network analysis. Each leaf on the dendrogram represents a single gene, and the color at the bottom indicates the co-expression module assignment.

(C) Top 5 key module eigengenes (kMEs) of genes in each module.

(D-F) A lollipop plot showing the differential module eigengene (DME) between pre- and post-treatment in all samples (D), samples with PFS > 1 year (E) and PFS < 1 year (F).

The size of each dot corresponds to the number of genes in that module. An “X” is placed over each point that does not reach statistical significance of Mann-Whitney U test.
(G-H) Dot plot showing the top 5 terms for one KEGG (G) and GO_Biological_Process (H) database in module Epi-M4 and M8.

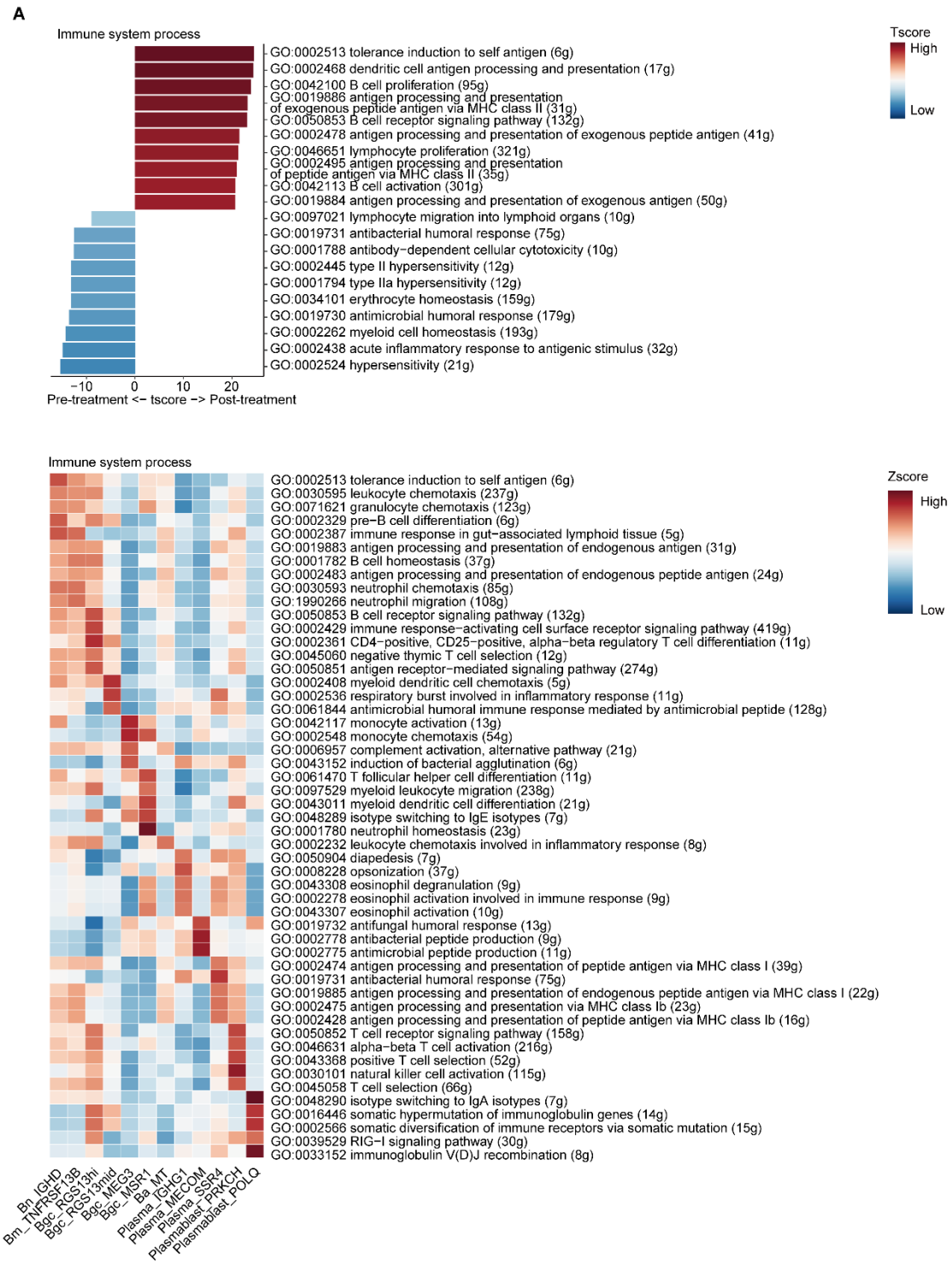


Supplementary Figure 11. Transcriptional UMAP Visualization of B cell and plasma Cell Subclusters Annotated by Sample, Time and Canonical Markers, related to Figure 5

(A) UMAP visualization of B cell and plasma cell subclusters, with sample annotated by sample.

(B) UMAP visualization of B cell and plasma cell subclusters, with sample annotated by sampling time.

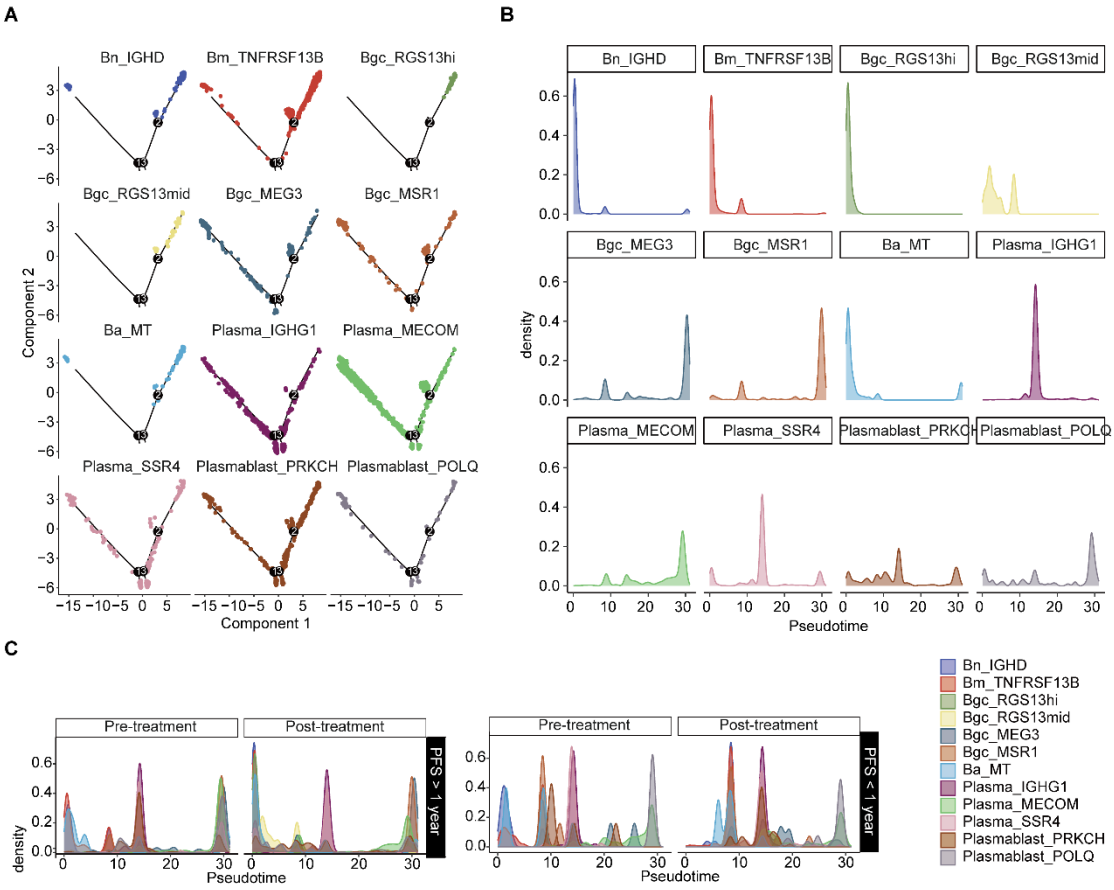
(C) UMAP visualization showing the expression of canonical marker genes across subtype.



Supplementary Figure 12. Characterization of B Subclusters, related to Figure 5

(A) The activities of immune system process in B cell and plasma cell between pre- and post-treatment.

(B) Heatmap showing the activities of immune system process in B cell and plasma cell subclusters. Colors indicate the Z score scaled value.

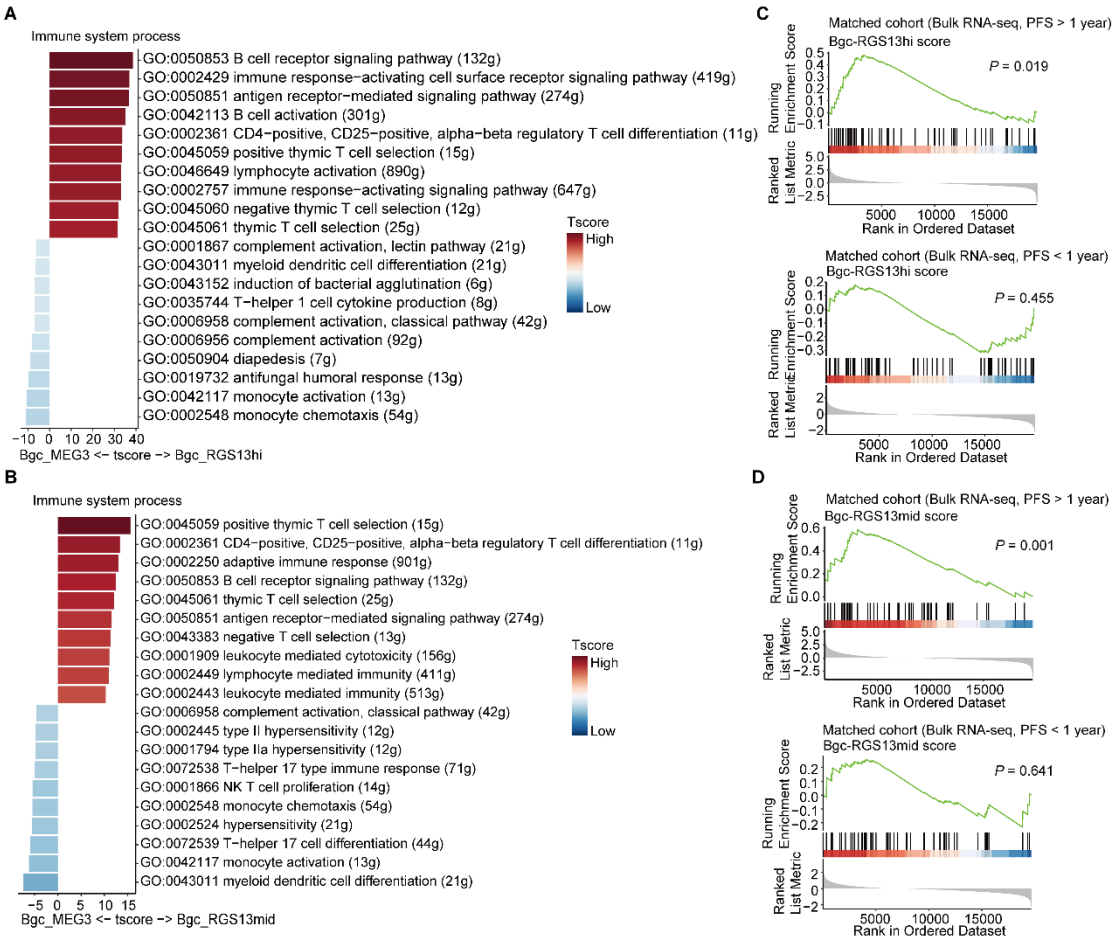


Supplementary Figure 13. Trajectory Analysis of B&Plasma, related to Figure 5

(A) Monocle2 trajectory analysis showing developmental trajectory across B cell and plasma cell subclusters, with cells colored and split by subclusters.

(B) Distribution of subclusters across pseudotime, colored and split by subclusters.

(C) Distribution of subclusters across pseudotime, colored by subclusters and split by sampling time and subgroup.



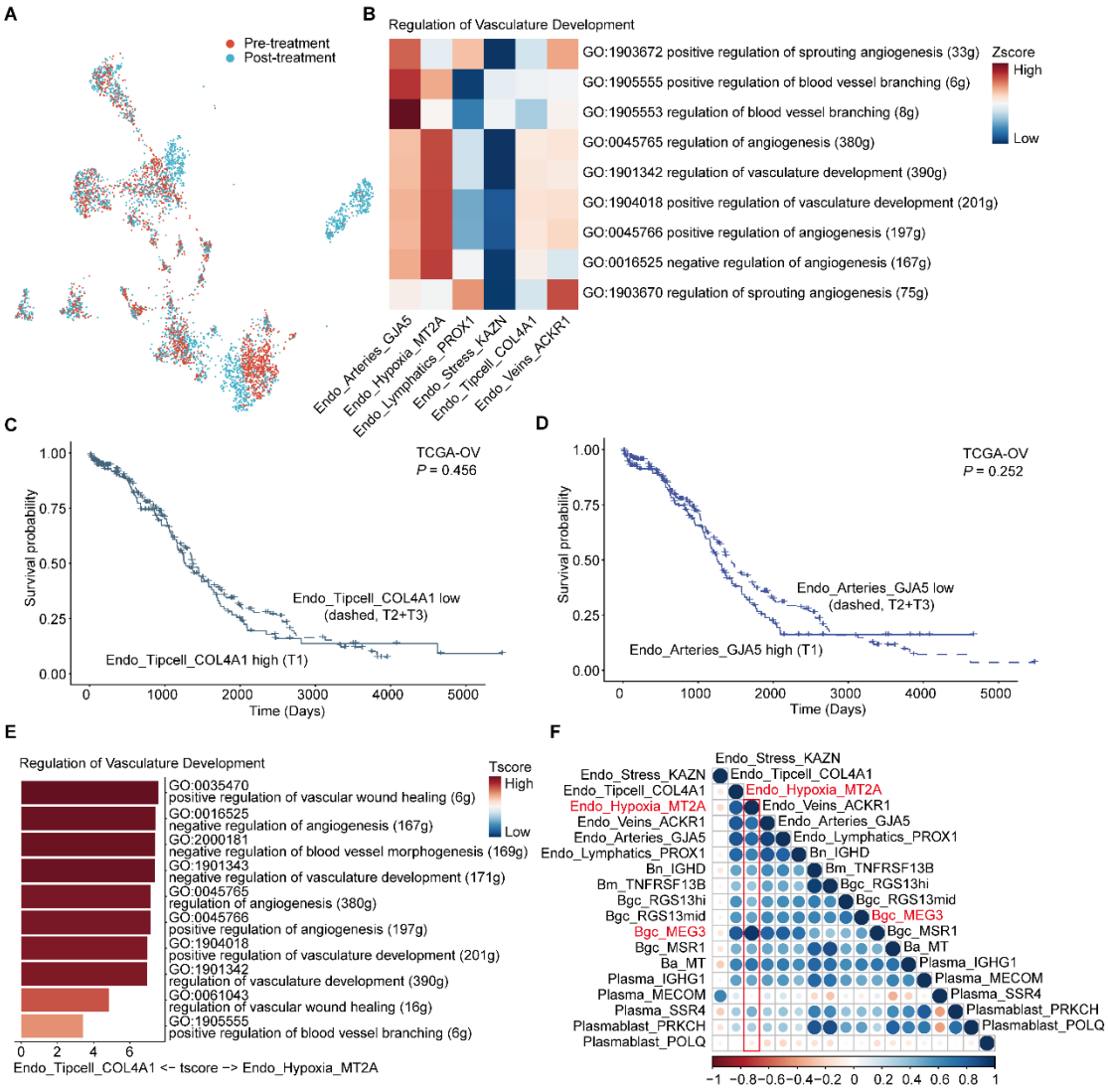
Supplementary Figure 14. Pathway Activity and Longitudinal Enrichment of Distinct B Subpopulation, related to Figure 5

(A) The activities in regulation of vasculature development between Bgc_MEG3 and Bgc_RGS13hi. Color indicates the T score scaled value.

(B) The activities in regulation of vasculature development between Bgc_MEG3 and Bgc_RGS13mid. Color indicates the T score scaled value.

(C) Bgc-RGS13hi score enrichment in post-treatment samples compared to pre-treatment samples of patients with PFS > 1 year (up) and PFS < 1 year (bottom).

(D) Bgc-RGS13mid score enrichment in post-treatment samples compared to pre-treatment samples of patients with PFS > 1 year (up) and PFS < 1 year (bottom).



Supplementary Figure 15. Characterization of Endothelial Subpopulations, related to Figure 6

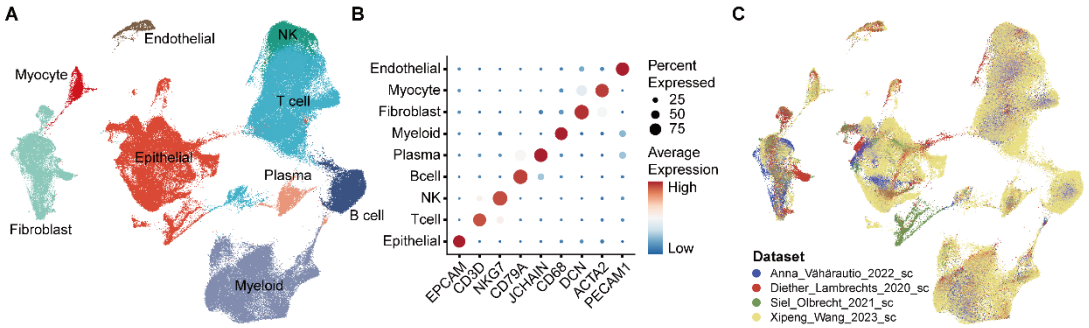
(A) UMAP visualization of endothelial subclusters, with sample annotated by sampling time.

(B) Heatmap showing the activities for regulation of vasculature development in endothelial subclusters. Colors indicate the Z score scaled value.

(C-D) The Kaplan-Meier curves comparing the probability of survival according to Endo_Tipcell_COL4A1 score (C) and Endo_Arteries_GJA5 score (D) in TCGA-OV cohort. Patients were dichotomized into two groups based on the score distribution: the high-score group comprised the top one-third of patients with the highest scores, while the low-score group included the remaining two-thirds. P values were calculated with log-rank test.

(E) The activities in regulation of vasculature development in between Endo_Tipcell_COL4A1 and Endo_Hypoxia_MT2A. Color indicates the T score scaled value.

(F) Corrplot showing pearson association between endothelial subtype and B subtype score in TCGA-OV cohort.



Supplementary Figure 16. Integrated Published Datasets, related to Figure 7

(A) UMAP visualization of scRNA-seq data from published datasets, with major cell types annotated by distinct colors.

(B) Dot plot showing the expression of canonical marker genes across major cell type.

(C) UMAP visualization of scRNA-seq data, with data source annotated by distinct colors.



Supplementary Figure 17. Spatial Distribution of Major Cell Types in P14 and P17, related to Figure 7

(A-B) Spatial distribution of major cell types inferred by CARD-based deconvolution of spatial transcriptomic data in pre- (A) and post-treatment (B) samples of patient P14.

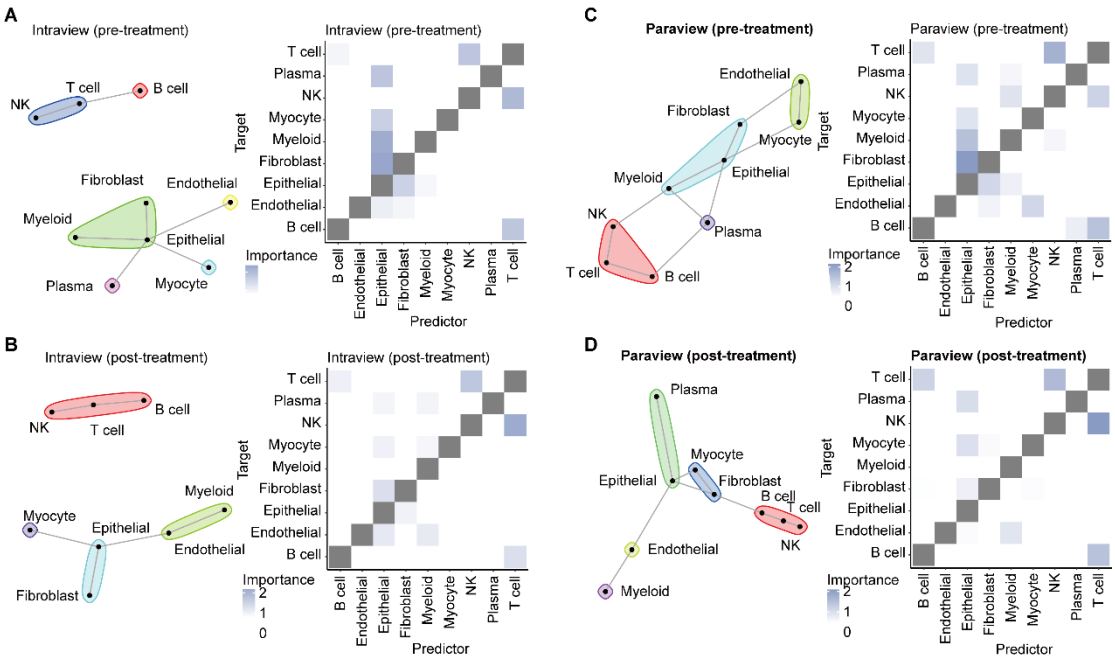
(C-D) Spatial distribution of major cell types inferred by CARD-based deconvolution of spatial transcriptomic data in pre- (C) and post-treatment (D) samples of patient P17.



Supplementary Figure 18. Spatial Distribution of Major Cell Types in P18 and P20, related to Figure 7

(A-B) Spatial distribution of major cell types inferred by CARD-based deconvolution of spatial transcriptomic data in pre- (A) and post-treatment (B) samples of patient P18.

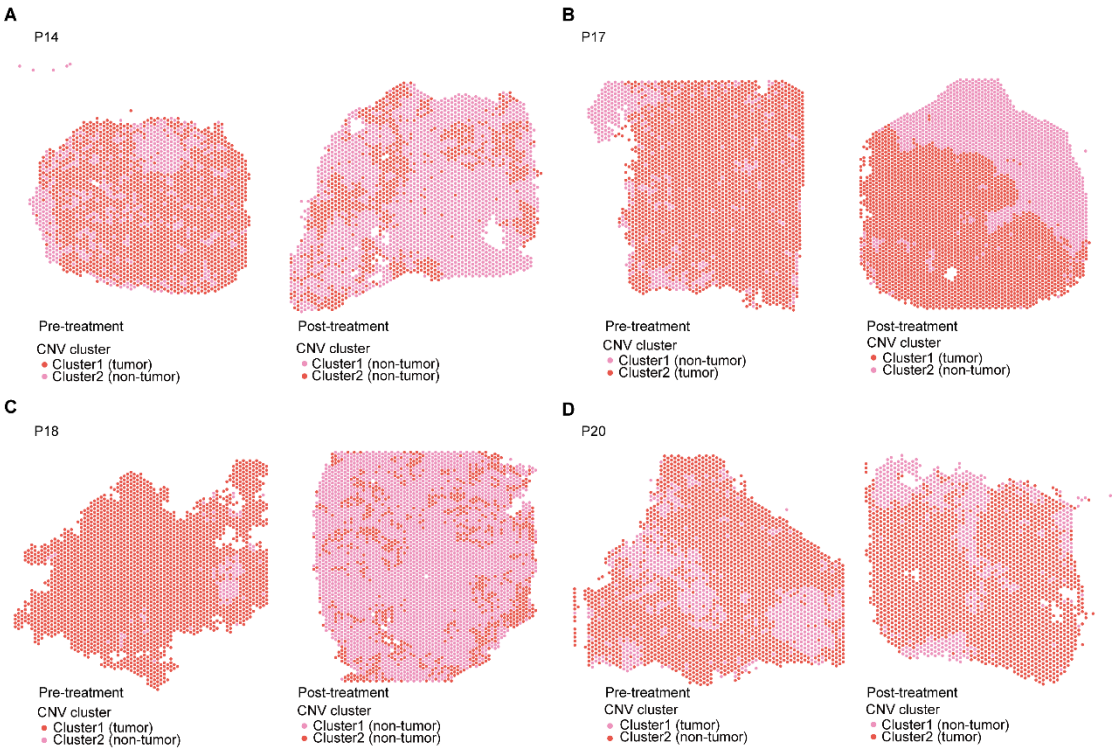
(C-D) Spatial distribution of major cell types inferred by CARD-based deconvolution of spatial transcriptomic data in pre- (C) and post-treatment (D) samples of patient P20.



Supplementary Figure 19. MistyR-derived Intraview and Paraview Model, related to Figure 7

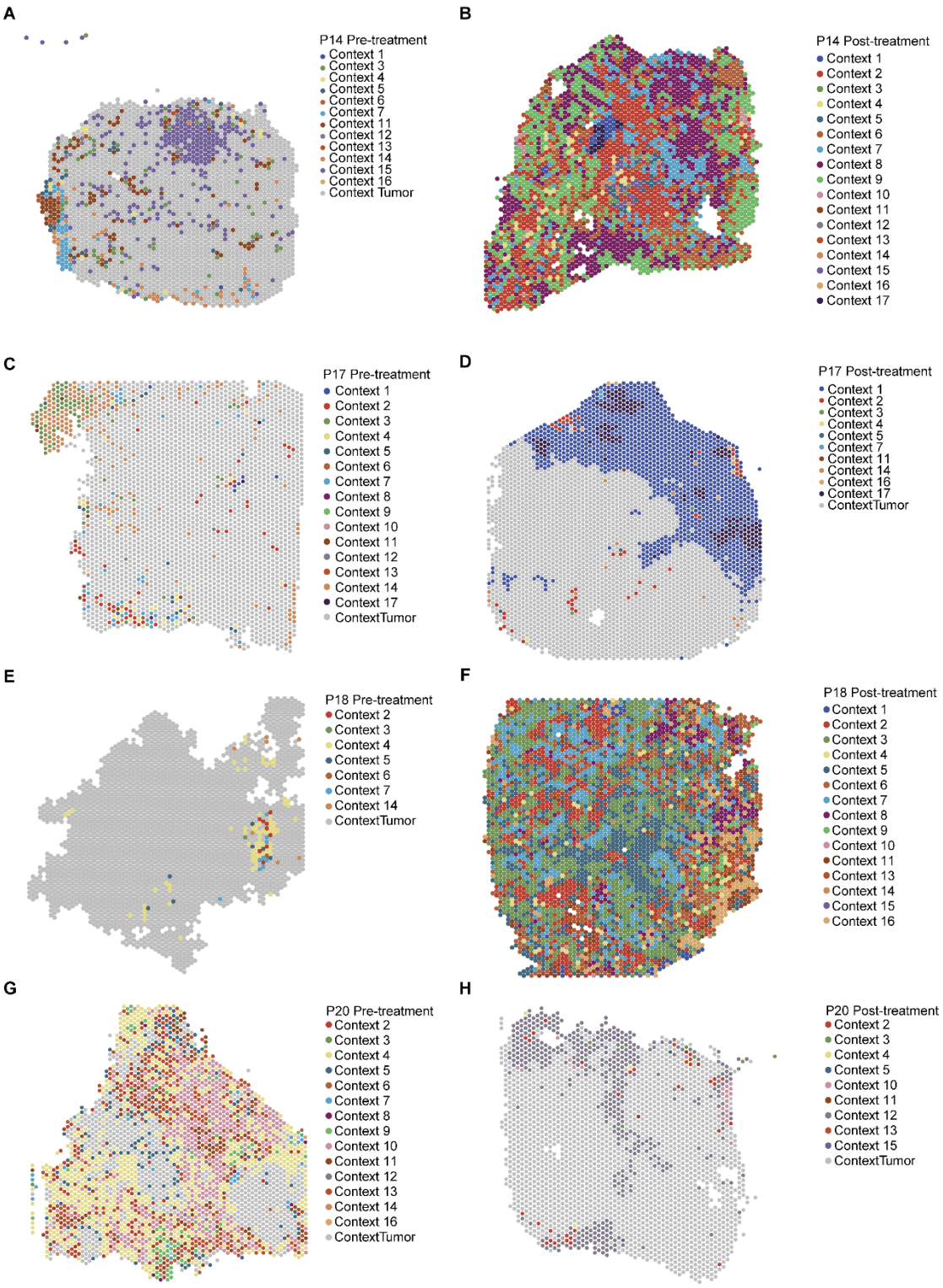
(A-B) MistyR-derived Intraview model in pre-treatment (A) and post-treatment (B) samples, illustrating spatial influences from neighboring tissue regions.

(C-D) MistyR-derived Paraview model in pre-treatment (C) and post-treatment (D) samples, illustrating spatial influences from neighboring tissue regions.



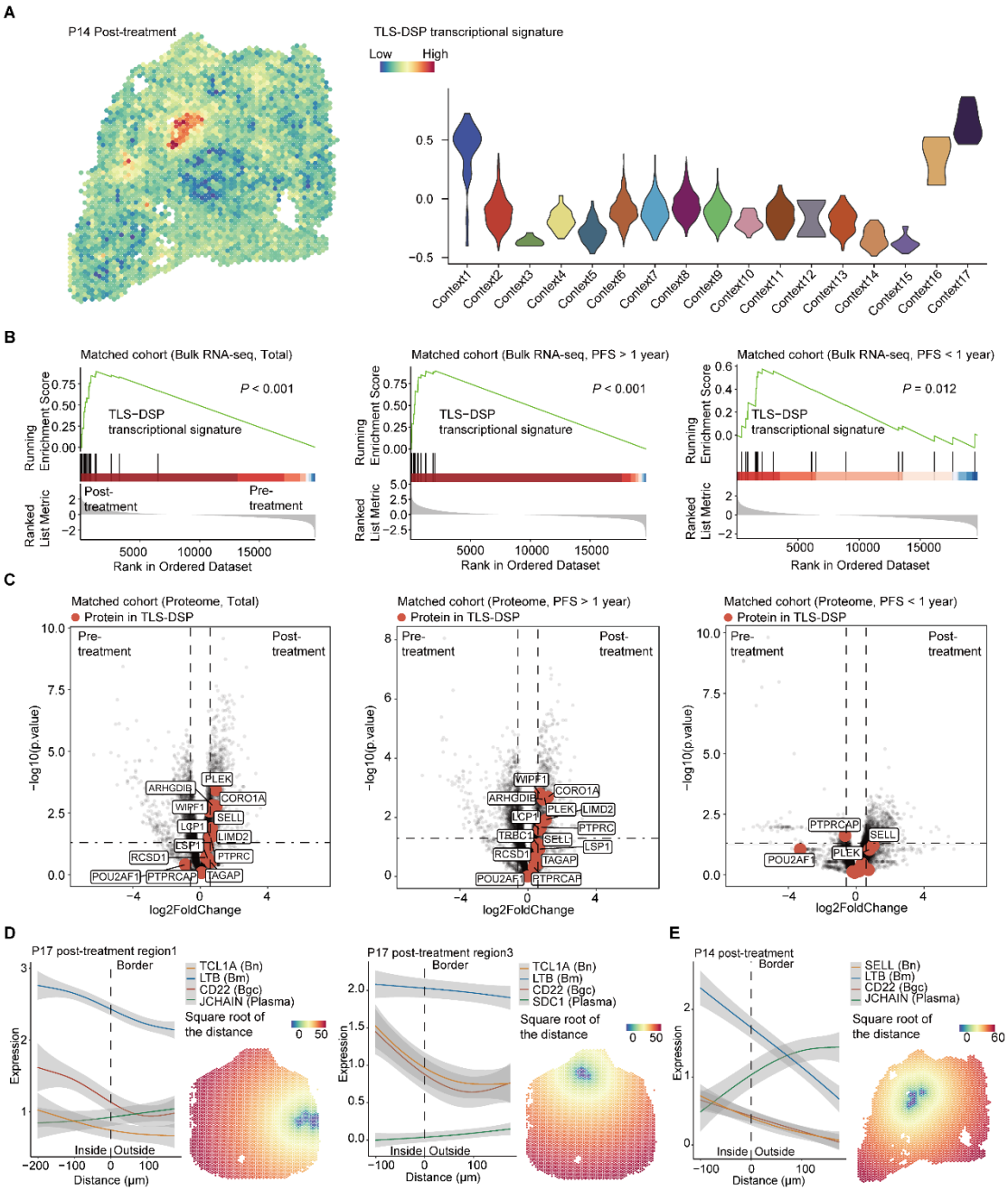
Supplementary Figure 20. Spatial Distribution of CNV cluster, related to Figure 7

(A-D) Spatial distribution of CNV cluster in pre (left)/post (right)-treatment sample of patient P14 (A), P17 (B), P18 (C) and P20 (D).



Supplementary Figure 21. Spatial Distribution of Spatial Context, related to Figure 7

(A-B) Spatial distribution of context in pre (a)/post (b)-treatment samples of patient P14.
(C-D) Spatial distribution of context in pre (c)/post (d)-treatment sample of patient P17.
(E-F) Spatial distribution of context in pre (e)/post (f)-treatment samples of patient P18.
(G-H) Spatial distribution of context in pre (g)/post (h)-treatment samples of patient P20.



Supplementary Figure 22. Spatial Localization, Transcriptional and Proteomic Features of TLS-DSP Signature, related to Figure 7

(A) Spatial distribution of TLS-DSP score in post-treatment sample of patient P14 (left) and violin plot showing distribution of TLS-DSP in spatial contexts.

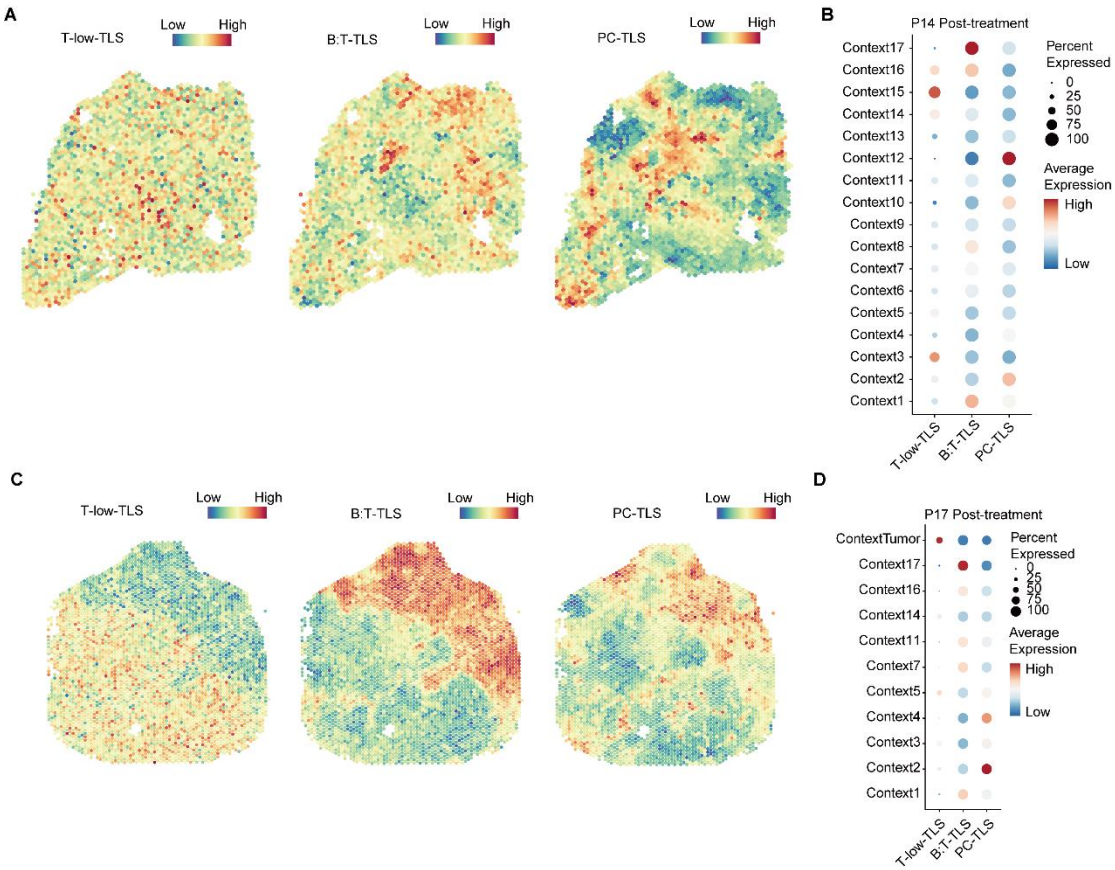
(B) TLS-DSP transcriptional signature enrichment in post-treatment samples compared to pre-treatment samples of all sample-matched patients (left), those with PFS > 1 year (middle) and PFS < 1 year (right).

(C) Volcano plot of proteomic data showing differentially expressed proteins between pre- and post-treatment samples in sample-matched patients (left), those with PFS > 1 year (middle) and PFS < 1 year (right), with TLS-DSP–related proteins annotated by color.

(D) Spatial expression pattern of B cell major lineages specific genes in region 1 and region 3 of post-treatment P17 sample. Spatial distribution of distance from region 1 and

region 3 of post-treatment P17 sample.

(E) Spatial expression pattern of B cell major lineages specific genes in region of post-treatment P14 sample (left). Spatial distribution of distance from region of post-treatment P14 sample (right).



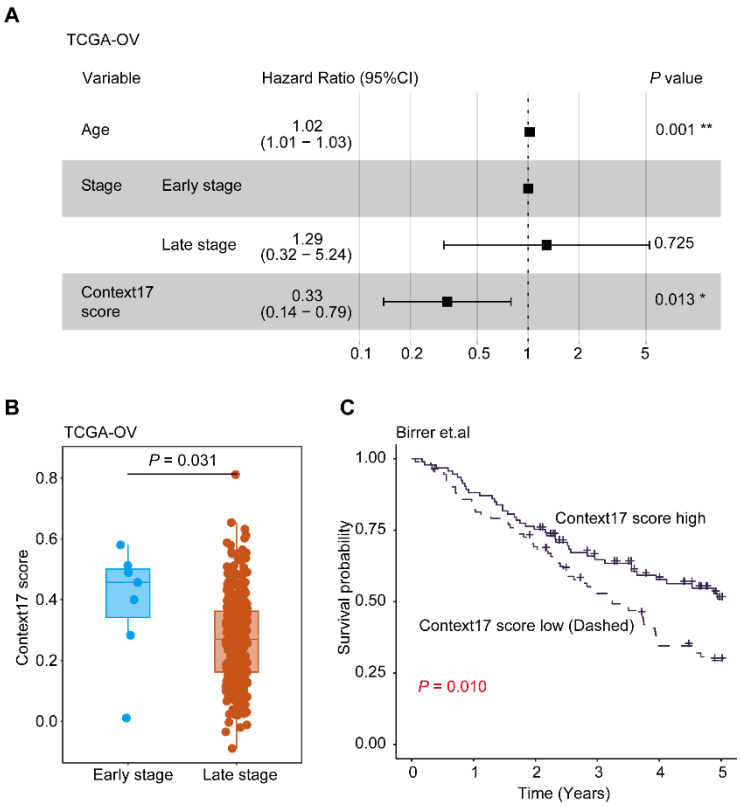
Supplementary Figure 23. Spatial Distribution of TLS Subtype Signature, related to Figure 7

(A) Spatial distribution of T-low-TLS (left), B:T-TLS (middle) and PC-TLS (right) signature in post-treatment sample of patient P14.

(B) Dot plot showing the expression pattern of T-low-TLS, B:T-TLS and PC-TLS signature across spatial contexts in post-treatment sample of patient P14.

(C) Spatial distribution of T-low-TLS (left), B:T-TLS (middle) and PC-TLS (right) signature in post-treatment sample of patient P17.

(D) Dot plot showing the expression pattern of T-low-TLS, B:T-TLS and PC-TLS signature across spatial contexts in post-treatment sample of patient P17.

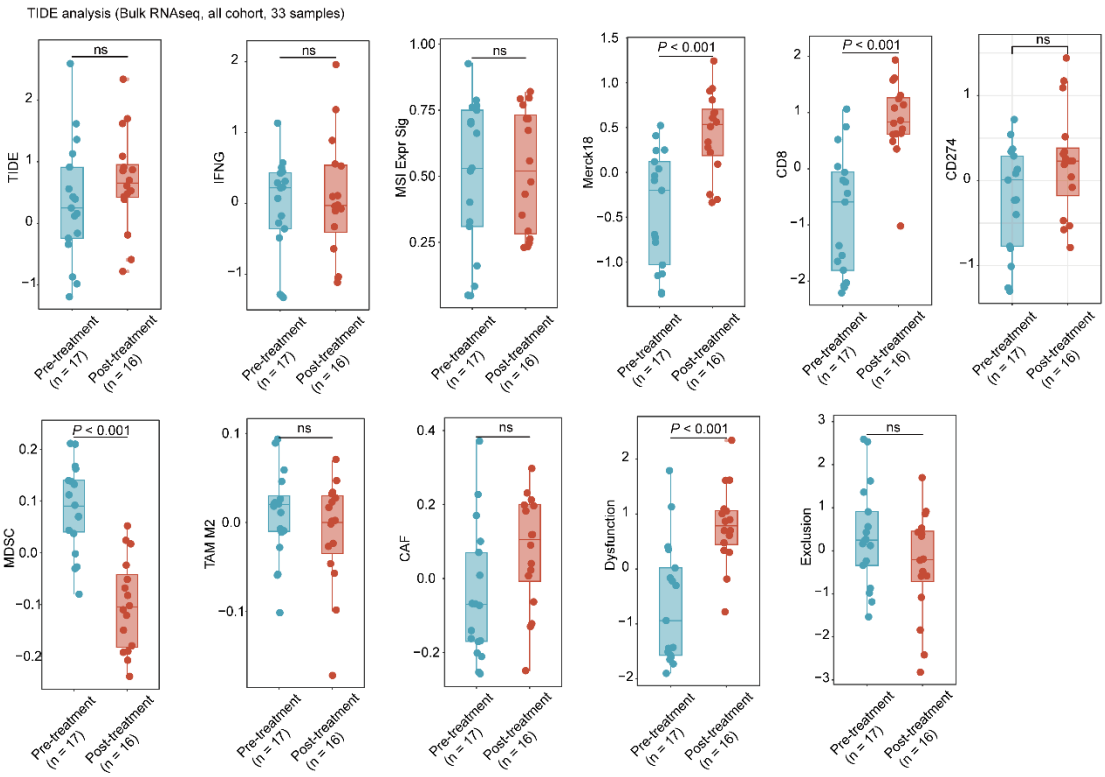


Supplementary Figure 24. Clinical Relevance of the Context 17 Score, related to Figure 7

(A) Forest plot showing multivariate Cox proportional hazards model of age, stage and Context 17 score.

(B) Boxplot showing Context 17 score between early and late stage in TCGA-OV cohort.

(C) The Kaplan-Meier curves comparing the probability of survival according to Context 16 score in Birrer et al. cohort. Patients are dichotomized as high or low score using the median Context 17 score as cutoff. P values were calculated with log-rank test.



Supplementary Figure 25. TIDE analysis of PASSION Cohort, related to Figure 8.
Box-and-scatter plots of TIDE and tumor microenvironment signature scores derived from bulk RNA-seq profiling of the full clinical cohort. Intergroup comparisons between pre- and post-treatment paired samples were performed using unpaired Student's t-test; ns, not significant